

# Discrete Sliding Mode Control with Actuator Constraints for the Stabilization of a Ballistically Launched Drone

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**Highly unstable initial states, such as those produced by ballistic deployment, require nonlinear controllers to stabilize the tumble before the vehicle crashes. We derive and test two nonlinear controllers for which we establish Lyapunov-based stability conditions: a continuous-time Sliding Mode Controller (cSMC) and a discrete-time Sliding Mode Controller (dSMC). These conditions govern the four actuated channels (altitude, roll, pitch, and yaw); the cSMC guarantee is continuous-time and does not survive discretization, whereas the dSMC is built directly in discrete time and holds under the discretized plant. We test these against two off-the-shelf controllers: the Linear Quadratic Regulator with Integral Action (LQR) and the cascaded Proportional-Integral-Derivative (PID) controller. Sampling 17 deployment states along one ballistic trajectory, we find the dSMC recovers more of them than the LQR, PID, or continuous-time controllers. We then extend the dSMC to account for possible motor control constraints in terms of maximum squared rotational speed. If the desired motor speeds fall within the physical bounds, then the augmented controller and the ideal controller are identical, and we satisfy the exponential reaching law. Otherwise, we saturate the control inputs in order of increasing importance—yaw first, then altitude/thrust, then pitch/roll—and obtain a weaker input-to-state stability (ISS) bound that holds while the trajectory stays within an explicit envelope. However, despite this weaker stability, on a separate 140-trial landing sweep the priority-weighted saturation raises average landing-site reachability roughly  $2.7\times$  over simple per-motor clipping, from 22.1% to 60.7%.**

## I. Introduction

In this paper, we develop a controller for UAV stabilization following ballistic deployment. Current work in ballistically launched UAVs, such as the Caltech SQUID [1] or the Texas A&M TLMAV [2], focuses on drones that transition into powered flight at a predetermined point in their ballistic trajectory. Current military research efforts, such as the Launched Effects program [3], call for a flexible platform that can be deployed anywhere, requires minimal infrastructure, and can carry arbitrary payloads. For example, quick-reaction drones could deliver medical supplies, a role examined in feasibility modeling studies [4]. Alternatively, launched drones could create a mobile dispersed sensor network, whether for wide-area surveying [5] or force-protection surveillance [6]. These applications require drones capable of deploying into powered flight at any point along their trajectory. We evaluate the controllers on these test cases by how much of the ballistic-deployment envelope each recovers.

### A. Related Work

The dSMC approach for quadrotor control has been the subject of some limited existing studies. Xiong and Zhang [7] introduced the unconstrained formulation of the dSMC and presented how a discrete Lyapunov stability condition may be satisfied for arbitrary discretization sampling times. Xu et al. [8] introduced per-channel auxiliary states for the three attitude moments of a quadrotor using SMC under input saturation, and proved closed-loop stability via Lyapunov analysis. Kuang and Chen [9] adopted a vector-valued second-order auxiliary controller for the position and attitude loops together with a disturbance observer and adaptive laws for unknown mass and inertia. Shao et al. [10], on the other hand, embedded a bounded saturation function directly in the control law alongside adaptive parameters. However, these works are all in the continuous time space, and do not cover the collective-thrust control channel together with the three torque channels in a single per-motor construction.

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While these saturation methods do not incorporate control prioritization, Faessler et al. [11] present an explicit prioritization of roll and pitch torques above collective thrust above yaw torque under motor saturation, and build the priority into an iterative mixer without formal stability assurances. Similarly, Mueller and D’Andrea [12] linearize their input constraints inside an MPC loop. We use the same priority order ( $M_x, M_y \succ T \succ M_z$ ) but solve the projection over the squared rotor speed  $\Omega^2$ -space and carry it through to the closed-loop stability proof.

In addition to retaining the basic Lyapunov stability bounds, Wang et al. [13] give us the discrete-time component-wise saturation model and the saturation-dependent Lyapunov approach we extend to characterize the admissible control space  $\mathcal{U}$ , and Jiang and Wang [14] provide the discrete-time input-to-state-stability (ISS) Lyapunov framework we use to demonstrate the ISS form of our closed-loop result. The discrete reaching-law boundary  $\eta \Delta t < 2$ , which follows from the discrete-time sliding-mode existence condition  $|s_{k+1}| < |s_k|$  of Sarpturk et al. [15], drops out as a special case when the deficit  $\Delta u_k$  vanishes; the unconstrained dSMC structure follows the discrete-time formulation of Xiong and Zhang [7].

## B. New Contributions

We present a corrected derivation with formal Lyapunov stability analysis of the Discrete Sliding Mode Controller (dSMC) presented in [7]. We also extend the dSMC to include per-motor speed limits, which equate to control effort saturation in collective thrust, and roll, pitch, and yaw moments. We compare this to a reformulated Continuous Sliding Mode Controller (cSMC), similar to the form derived in [16]. We compare these two SMC formulations against classical LQR and cascaded PID, both of which are common commercial off-the-shelf (COTS) drone control solutions. Our dSMC corrects and re-derives the control laws in [7], whose proposed forms violate the exponential reaching condition and produce an unstable or unresponsive controller. We also introduce a new test case, the stabilization of a ballistically launched quadrotor into powered flight, which suggests that the results of [16] hold only for a limited range of initial conditions. Lastly, we reformulate the dSMC controller with control saturation. We first apply control-priority weighting (with pitch and roll weighted as more important than total thrust, which in turn is weighted as more important than yaw). Then, we augment each control channel with auxiliary states and, once again, apply the exponential reaching law analysis, demonstrating that the reaching laws hold under transient control saturation. When commands are naturally feasible, we recover the unconstrained dSMC controller. In the presence of control saturation, we demonstrate an input-to-state stability bound (provided the saturation is not permanent). This approach is similar to [13, 14], but we adapt these methods to discrete time control with a separate scalar auxiliary state per actuator channel (including total thrust) and show that we still maintain the Lyapunov bound of the underlying dSMC under transient saturation. We also examine a new application of the controller; ballistic transition stabilization and landing.

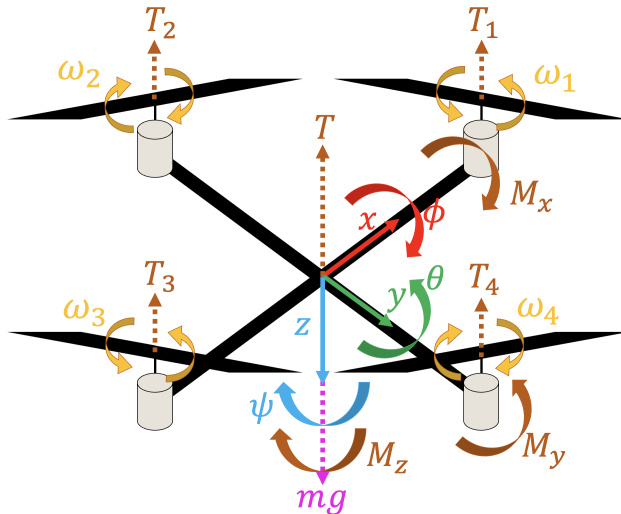


Fig. 1 Quadrotor coordinate system and force vectors.

**Table 1 Nomenclature and Variables**

| Symbol                                     | Description                                                              | Value                                            |
|--------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------|
| <b>Model parameters</b>                    |                                                                          |                                                  |
| $k_{MT}$                                   | Ratio of motor torque to thrust                                          | 0.1 m                                            |
| $l$                                        | Quadrotor arm length                                                     | 0.2 m                                            |
| $J_{xx}$                                   | Moment of inertia about body $x$ axis                                    | $1.8 \times 10^{-3} \text{ kg} \cdot \text{m}^2$ |
| $J_{yy}$                                   | Moment of inertia about body $y$ axis                                    | $1.8 \times 10^{-3} \text{ kg} \cdot \text{m}^2$ |
| $J_{zz}$                                   | Moment of inertia about body $z$ axis                                    | $1.5 \times 10^{-3} \text{ kg} \cdot \text{m}^2$ |
| $m$                                        | Drone mass                                                               | 0.8 kg                                           |
| $g$                                        | Acceleration due to gravity                                              | $9.81 \text{ m/s}^2$                             |
| $J_r$                                      | Rotor moment of inertia about the spin axis of the motor                 | $\approx 0$                                      |
| $\kappa$                                   | Gyroscopic coupling between rotors and body                              | $\approx 0$ (see text)                           |
| <b>State variables</b>                     |                                                                          |                                                  |
| $v_x, v_y, v_z$                            | Body-frame velocity components                                           | —                                                |
| $\omega_x, \omega_y, \omega_z$             | Body-frame rotation rates                                                | —                                                |
| $\theta, \phi, \psi$                       | Pitch, roll, and yaw Euler angles (Z-Y-X order)                          | —                                                |
| $x, y, z$                                  | Inertial position; north, east positive; $z$ upward (altitude)           | —                                                |
| <b>Motor mixing</b>                        |                                                                          |                                                  |
| $T$                                        | Net thrust along the body $z$ axis                                       | —                                                |
| $M_x, M_y, M_z$                            | Moments about the body $x, y, z$ axes                                    | —                                                |
| $T_i$                                      | Thrust produced by rotor $i, i \in \{1, \dots, 4\}$                      | —                                                |
| $\omega_i$                                 | Angular velocity of rotor $i$                                            | —                                                |
| $\Omega_i^2$                               | Squared angular velocity of rotor $i$                                    | —                                                |
| $\Omega_r$                                 | Net rotor speed differential $\omega_1 - \omega_2 + \omega_3 - \omega_4$ | —                                                |
| $\Omega_{\min}^2, \Omega_{\max}^2$         | Per-rotor squared-speed bounds                                           | 0, 2 (normalized)                                |
| $T_M$                                      | Mixing matrix from $\Omega^2$ to $(T, M_x, M_y, M_z)^T$                  | —                                                |
| $b$                                        | Per-rotor thrust per squared rotor speed                                 | 5 (normalized)                                   |
| $d$                                        | Per-rotor yaw torque per squared rotor speed                             | 2 (normalized)                                   |
| $u_k$                                      | Commanded control at step $k, (T, M_x, M_y, M_z)^T$                      | —                                                |
| $\bar{u}_k$                                | Saturated command after projection onto $\mathcal{U}$                    | —                                                |
| $\Delta u_k$                               | Saturation deficit, $u_k - \bar{u}_k$                                    | —                                                |
| $\mathcal{U}$                              | Feasible control set under per-rotor speed limits                        | —                                                |
| <b>Sliding mode control and saturation</b> |                                                                          |                                                  |
| $s_\alpha$                                 | Sliding surface for channel $\alpha \in \{T, M_x, M_y, M_z\}$            | —                                                |
| $\tilde{s}_k$                              | Modified sliding surface, $s_k - D_k \xi_k$                              | —                                                |
| $e_\alpha$                                 | Tracking error for channel $\alpha$                                      | —                                                |
| $\eta_\alpha$                              | Reaching-law gain for channel $\alpha$                                   | —                                                |
| $\xi_k$                                    | Per-channel auxiliary state vector                                       | —                                                |
| $k_{\xi, \alpha}$                          | Contraction rate of the auxiliary-state filter                           | —                                                |
| $D_k$                                      | Coupling between the auxiliary state and the sliding surface             | —                                                |
| $V_k$                                      | Discrete Lyapunov function at step $k$                                   | —                                                |
| <b>Model discretization</b>                |                                                                          |                                                  |
| $\Delta t$                                 | Discrete control sample interval                                         | 0.02 s                                           |
| $k$                                        | Discrete-time index                                                      | —                                                |

## II. Vehicle Model

We apply the controllers derived in this paper to the standard symmetric quadrotor (Figure 1) in a North-East-Down inertial coordinate frame with Z-Y-X Euler angle rotation order. We use the standard nonlinear and linear system of equations presented in [16]. The model parameters are given in Table 1. We choose to work in the force-moment space ( $T$  for net thrust along the body  $z$  axis and  $M_x, M_y, M_z$  for the moments about the body  $x, y$ , and  $z$  axes), rather than the individual rotor angular velocities ( $\omega_1, \omega_2, \omega_3, \omega_4$ ), without loss of generality due to the linear mapping from squared rotor velocities to force-moments.

$$\begin{bmatrix} T \\ M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ k_{MT} & -k_{MT} & k_{MT} & -k_{MT} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix}. \quad (1)$$

The full nonlinear system of equations of motion is given in Equation 2, where  $v_x, v_y, v_z$  are the body-frame velocity components,  $\omega_x, \omega_y, \omega_z$  are the body-frame attitude rates,  $\theta, \phi, \psi$  are the inertial pitch, roll, and yaw Euler angles, and  $x, y, z$  are the inertial position (north and east positive; the  $z$  position is expressed as altitude, positive upward).

$$\begin{aligned} \dot{v}_x &= -v_z \omega_y + v_y \omega_z - g \sin \theta, & \dot{v}_y &= -v_x \omega_z + v_z \omega_x + g \cos \theta \sin \phi, & \dot{v}_z &= -v_y \omega_x + v_x \omega_y + g \cos \theta \cos \phi - \frac{T}{m}, \\ \dot{\omega}_x &= \frac{1}{J_{xx}} [-\omega_y \omega_z (J_{zz} - J_{yy}) + M_x - \kappa M_z \omega_y], & \dot{\omega}_y &= \frac{1}{J_{yy}} [-\omega_x \omega_z (J_{xx} - J_{zz}) + M_y - \kappa M_z \omega_x], & \dot{\omega}_z &= \frac{M_z}{J_{zz}}, \\ \dot{\theta} &= \omega_y \cos \phi - \omega_z \sin \phi, & \dot{\phi} &= \omega_x + \omega_y \sin \theta \tan \phi + \omega_z \cos \theta \tan \phi, & \dot{\psi} &= \frac{\omega_y \sin \phi + \omega_z \cos \phi}{\cos \theta}, \\ \dot{x} &= v_x \cos \psi \cos \theta + v_y (-\sin \psi \cos \phi + \cos \psi \sin \theta \sin \phi) + v_z (\sin \psi \sin \phi + \cos \psi \sin \theta \cos \phi), & \dot{y} &= v_x \sin \psi \cos \theta + v_y (\cos \psi \cos \phi + \sin \psi \sin \theta \sin \phi) + v_z (-\cos \psi \sin \phi + \sin \psi \sin \theta \cos \phi), & \dot{z} &= v_x \sin \theta - v_y \cos \theta \sin \phi - v_z \cos \theta \cos \phi. \end{aligned} \quad (2)$$

In the simulations reported here the rotor-body gyroscopic terms are inactive: the rotor spin inertia is negligible ( $J_r \approx 0$ ), and the  $\kappa M_z \omega_x, \kappa M_z \omega_y$  coupling terms are disabled accordingly ( $\kappa \rightarrow 0$ ); both are retained in the equations above for generality.

Linearizing the system about the steady-state point (level and stationary) yields a simplified set of equations that we use to derive our LQR controller gains.

$$\begin{aligned} \dot{v}_x &= -g \theta, & \dot{v}_y &= g \phi, & \dot{v}_z &= -T/m, \\ \dot{\omega}_x &= M_x/J_{xx}, & \dot{\omega}_y &= M_y/J_{yy}, & \dot{\omega}_z &= M_z/J_{zz}, \\ \dot{\theta} &= \omega_y, & \dot{\phi} &= \omega_x, & \dot{\psi} &= \omega_z, \\ \dot{x} &= v_x, & \dot{y} &= v_y, & \dot{z} &= -v_z. \end{aligned} \quad (3)$$

We skip the controllability and observability analysis. The quadrotor is well known to be fully (directly or indirectly) controllable in the nonlinear form above. We assume the full state is observable; on our hardware target, GPS, an onboard IMU, and an EKF provide position, attitude, and translational and rotational velocities with sufficient precision.

## III. Controller Formulation

### A. LQR Controller with Integral Action

LQR computes a constant state-feedback gain that minimizes an infinite-horizon quadratic cost. The cost function we minimize, given in Equation 4, is the standard form of the LQR cost function with integral action.

$$\begin{aligned} J(u) &= \int_0^\infty (\tilde{x}^\top Q \tilde{x} + u^\top R u) dt \\ \tilde{x} &= \left[ \left( \int_0^t e(\tau) d\tau \right)^\top \quad x^\top \right]^\top \\ e &= r - [x, y, z]^\top, \quad u = \frac{K_i}{s} e - K_p x \end{aligned} \quad (4)$$

**Table 2** PID gains for outer/inner position and attitude loops.

| Loop             | Channel                                  | P    | I | D    | N   |
|------------------|------------------------------------------|------|---|------|-----|
| Outer positional | $x \rightarrow \dot{x}$                  | 0.25 | 0 | 0    | –   |
|                  | $y \rightarrow \dot{y}$                  | 0.25 | 0 | 0    | –   |
|                  | $z \rightarrow \dot{z}$                  | 0.40 | 0 | 0    | –   |
| Inner positional | $\dot{x} \rightarrow \ddot{x}$           | 0.30 | 0 | 0.20 | 100 |
|                  | $\dot{y} \rightarrow \ddot{y}$           | 0.30 | 0 | 0.20 | 100 |
|                  | $\dot{z} \rightarrow \ddot{z}$           | 0.80 | 0 | 0    | –   |
| Outer attitude   | $\theta \rightarrow \dot{\theta}$        | 0.30 | 0 | 0    | –   |
|                  | $\phi \rightarrow \dot{\phi}$            | 0.30 | 0 | 0    | –   |
|                  | $\psi \rightarrow \dot{\psi}$            | 0.30 | 0 | 0    | –   |
| Inner attitude   | $\dot{\theta} \rightarrow \ddot{\theta}$ | 0.15 | 0 | 0    | –   |
|                  | $\dot{\phi} \rightarrow \ddot{\phi}$     | 0.15 | 0 | 0    | –   |
|                  | $\dot{\psi} \rightarrow \ddot{\psi}$     | 0.15 | 0 | 0    | –   |

**Table 3** PD controller gains for outer/inner position loops.

| Loop             | Channel                        | P | D | N  |
|------------------|--------------------------------|---|---|----|
| Outer positional | $x \rightarrow \dot{x}$        | 1 | 6 | 50 |
|                  | $y \rightarrow \dot{y}$        | 1 | 6 | 50 |
|                  | $z \rightarrow \dot{z}$        | 2 | 0 | –  |
| Inner positional | $\dot{x} \rightarrow \ddot{x}$ | 1 | 1 | 50 |
|                  | $\dot{y} \rightarrow \ddot{y}$ | 1 | 1 | 50 |
|                  | $\dot{z} \rightarrow \ddot{z}$ | 1 | 0 | –  |

To realize a LQR controller with integral action, we augment our base state vector with three additional states, representing integrated Earth-frame position. This way, we can derive integral gains in addition to proportional gains when we solve the Discrete Algebraic Riccati Equation. We use the following state-cost matrix  $Q$  and control-cost matrix  $R$ , which are the result of manual tuning of the weights presented in [16]:

$$Q = \text{diag}(q)/|q|_2, \quad R = I_4, \quad (5)$$

$$q = [3, 3, 6000, 1080, 1080, 1080, 180, 180, 180, 0.5, 0.5, 1000, 15, 15, 300]^\top.$$

Solving for the optimal proportional and integral gains, we obtain:

$$K_i = \begin{bmatrix} 0 & 0 & 0.22 \\ 0 & 0.05 & 0 \\ -0.05 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (6)$$

$$K_p = \begin{bmatrix} 0 & 0 & -1.55 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.91 \\ 0 & 0.16 & 0 & 0.42 & 0 & 0 & 0 & 1.56 & 0 & 0 & 0.13 & 0 \\ -0.16 & 0 & 0 & 0 & 0.42 & 0 & 1.16 & 0 & 0 & -0.13 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.41 & 0 & 0 & 0.17 & 0 & 0 & 0 \end{bmatrix}$$

### B. Cascaded PID Controller

We formulate the standard four-loop cascaded PID controller, with outer and inner loops for both position and attitude. The gains used are given in Table 2.

### C. Continuous Sliding Mode Controller

The cSMC is a Lyapunov-stable controller that directly accounts for the system's nonlinear dynamics. We follow the control laws of [16] with one modification. Where [16] uses a single PD layer to produce the desired Euler angles, we use two discrete PD layers: the first computes a desired translational acceleration, and the second resolves the corresponding attitude reference. This uses the cascaded structure used by most COTS flight controllers, and lets us tune the per-axis gains independently. The position PD control layer gains are listed in Table 3.

$$\alpha \in \{T, M_x, M_y, M_z\}, \quad s_\alpha = c_\alpha e_\alpha + \dot{e}_\alpha, \quad (7)$$

$$r_{PID} = \begin{bmatrix} \tilde{z} \\ \tilde{\phi} \\ \tilde{\theta} \\ \tilde{\psi} \end{bmatrix}, \quad e = r_{PID} - \begin{bmatrix} z \\ \phi \\ \theta \\ \psi \end{bmatrix}, \quad \begin{bmatrix} c_T \\ c_{M_x} \\ c_{M_y} \\ c_{M_z} \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 10 \\ 0.5 \end{bmatrix}.$$

$$\begin{aligned} \hat{u}_T = & -\frac{m c_T}{c \theta c \phi} (v_x s \theta - v_y c \theta s \phi - v_z c \theta c \phi) \\ & + m(-v_y \omega_x + v_x \omega_y + g c \theta c \phi) \end{aligned} \quad (8)$$

$$\hat{u}_{M_x} = -c_{M_x} J_{xx} (\omega_x + \omega_y s \phi t \theta + \omega_z c \phi t \theta) + \omega_y \omega_z (J_{zz} - J_{yy}) \quad (9)$$

$$\hat{u}_{M_y} = -\frac{c_{M_y} J_{yy}}{c \phi} (\omega_y c \phi - \omega_z s \phi) + \omega_x \omega_z (J_{xx} - J_{zz}) \quad (10)$$

$$\hat{u}_{M_z} = -c_{M_z} J_{zz} (\omega_y t \phi + \omega_z) \quad (11)$$

$$u_\alpha = \hat{u}_\alpha - K_\alpha \text{sat}(2s_\alpha, -1, 1) \quad (12)$$

$$\begin{aligned} K_T &= \frac{\gamma_T m}{c \theta c \phi}, \quad K_{M_x} = \gamma_{M_x} J_{xx}, \\ K_{M_y} &= \frac{\gamma_{M_y} J_{yy}}{c \phi}, \quad K_{M_z} = \frac{\gamma_{M_z} J_{zz} c \theta}{c \phi}, \\ \gamma_T &= 12, \quad \gamma_{M_x} = 8, \quad \gamma_{M_y} = 8, \quad \gamma_{M_z} = 1. \end{aligned} \quad (13)$$

We now present a brief Lyapunov stability proof for the cSMC formulation. We define the Lyapunov function  $V_\alpha(x)$  and its time derivative for each control channel  $\alpha \in \{T, M_x, M_y, M_z\}$ :

$$V_\alpha(x) = \frac{1}{2} s_\alpha^2, \quad \dot{V}_\alpha(x) = s_\alpha \dot{s}_\alpha. \quad (14)$$

The goal is to show that if  $s_\alpha$  stably converges to zero, the tracking error of the controller likewise converges to zero. Since  $\dot{s}_\alpha$  is the error response of the system (from Equation 7), we derive  $\dot{s}_\alpha$  directly from the nonlinear equations of motion (Equation 2):

$$\begin{aligned} \dot{s}_T &= c_T (v_x s \theta - v_y c \theta s \phi - v_z c \theta c \phi) \\ &\quad - \left( -v_y \omega_x + v_x \omega_y + g c \theta c \phi - \frac{u_T}{m} \right) c \theta c \phi, \\ \dot{s}_{M_x} &= c_{M_x} (\omega_x + \omega_y s \phi t \theta + \omega_z c \phi t \theta) \\ &\quad - \frac{1}{J_{xx}} (\omega_y \omega_z (J_{zz} - J_{yy}) - u_{M_x}), \\ \dot{s}_{M_y} &= c_{M_y} (\omega_y c \phi - \omega_z s \phi) - \frac{c \phi}{J_{yy}} (\omega_x \omega_z (J_{xx} - J_{zz}) - u_{M_y}), \\ \dot{s}_{M_z} &= c_{M_z} \left( \frac{\omega_y s \phi}{c \theta} + \frac{\omega_z c \phi}{c \theta} \right) + \frac{u_{M_z}}{J_{zz}} \frac{c \phi}{c \theta}. \end{aligned} \quad (15)$$

Substituting the control efforts from Equations 8–13, with  $K_\alpha$  given in their  $\gamma_\alpha$  form:

$$\begin{aligned}
\dot{s}_T &= c_T(v_x s_\theta - v_y c \theta s_\phi - v_z c \theta c \phi) \\
&\quad - \left[ (-v_y \omega_x + v_x \omega_y + g c \theta c \phi) - \frac{1}{m} \left( -\frac{mc_T}{c \theta c \phi} (v_x s_\theta - v_y c \theta s_\phi - v_z c \theta c \phi) \right. \right. \\
&\quad \left. \left. + m(-v_y \omega_x + v_x \omega_y + g c \theta c \phi) - \frac{\gamma_T m}{c \theta c \phi} \text{sat}(2s_T, -1, 1) \right) \right] c \theta c \phi, \\
\dot{s}_{M_x} &= c_{M_x}(\omega_x + \omega_y s_\phi t \theta + \omega_z c \phi t \theta) \\
&\quad - \frac{1}{J_{xx}} \left[ \omega_y \omega_z (J_{zz} - J_{yy}) - (-c_{M_x} J_{xx} (\omega_x + \omega_y s_\phi t \theta + \omega_z c \phi t \theta) \right. \\
&\quad \left. + \omega_y \omega_z (J_{zz} - J_{yy}) - \gamma_{M_x} J_{xx} \text{sat}(2s_{M_x}, -1, 1) \right), \\
\dot{s}_{M_y} &= c_{M_y}(\omega_y c \phi - \omega_z s \phi) \\
&\quad - \frac{c \phi}{J_{yy}} \left[ \omega_x \omega_z (J_{xx} - J_{zz}) - \left( -\frac{c_{M_y} J_{yy}}{c \phi} (\omega_y c \phi - \omega_z s \phi) + \omega_x \omega_z (J_{xx} - J_{zz}) \right. \right. \\
&\quad \left. \left. - \frac{\gamma_{M_y} J_{yy}}{c \phi} \text{sat}(2s_{M_y}, -1, 1) \right) \right], \\
\dot{s}_{M_z} &= c_{M_z} \left( \frac{\omega_y s \phi}{c \theta} + \frac{\omega_z c \phi}{c \theta} \right) \\
&\quad + \left( \frac{-c_{M_z} J_{zz} (\omega_y t \phi + \omega_z) - \frac{\gamma_{M_z} J_{zz} c \theta}{c \phi} \text{sat}(2s_{M_z}, -1, 1)}{J_{zz}} \right) \frac{c \phi}{c \theta}.
\end{aligned} \tag{16}$$

Equation 16 can be simplified to:

$$\begin{bmatrix} \dot{s}_T \\ \dot{s}_{M_x} \\ \dot{s}_{M_y} \\ \dot{s}_{M_z} \end{bmatrix} = - \begin{bmatrix} \gamma_T \text{sat}(2s_T, -1, 1) \\ \gamma_{M_x} \text{sat}(2s_{M_x}, -1, 1) \\ \gamma_{M_y} \text{sat}(2s_{M_y}, -1, 1) \\ \gamma_{M_z} \text{sat}(2s_{M_z}, -1, 1) \end{bmatrix}. \tag{17}$$

Choosing  $\gamma_\alpha > 0$ , the Lyapunov derivative becomes

$$\dot{V}_\alpha(x) = -\gamma_\alpha s_\alpha \text{sat}(2s_\alpha, -1, 1) < 0 \quad \forall s_\alpha \neq 0. \tag{18}$$

So,  $\dot{V}_\alpha(x)$  is negative definite,  $V_\alpha(x)$  is positive definite, and  $V_\alpha(0) = 0$ , and the Lyapunov stability requirements are satisfied for  $V_\alpha$  in Equation 14. The surfaces  $s_\alpha$  thus decay to zero over time, which, from Equation 7, is equivalent to the tracking error decaying to zero. The certified error  $e_\alpha$  is that of the four actuated channels ( $z, \phi, \theta, \psi$ ), with the outer reference  $r_{PID}$  held fixed while differentiating  $s_\alpha$ ; horizontal ( $x, y$ ) tracking is delegated to the outer position loop and is not covered by this inner-loop certificate. Additionally, within the boundary layer ( $|s_\alpha| < 1/2$ ),  $\text{sat}(2s_\alpha, -1, 1) = 2s_\alpha$ , and so

$$\begin{aligned}
\dot{V}_\alpha(x) &= -2\gamma_\alpha s_\alpha^2 = -4\gamma_\alpha \left( \frac{1}{2} s_\alpha^2 \right) \\
&= -\rho V_\alpha(x), \quad \rho = 4\gamma_\alpha > 0.
\end{aligned} \tag{19}$$

Outside the boundary layer,  $\text{sat}(2s_\alpha, -1, 1) = \text{sgn}(s_\alpha)$ , and  $\dot{V}_\alpha \leq -\gamma_\alpha |s_\alpha| < 0$ , driving the system into the boundary layer in finite time. Combining the two regimes shows that the tracking error is asymptotically stable, with an exponential rate of convergence to the sliding surface. ■

#### D. Discrete Sliding Mode Controller

The continuous dynamics assumption is strong. Flight computers and motor speed controllers are both discrete, and have varying duty cycles. We impose arbitrary 50 Hz zero-order holds on the input and output of the flight controller, and parameterize the discretization to be adjustable depending on the actual discretization of a given system. This violates the continuous-dynamics assumption of [16], and we must derive the dSMC to recover Lyapunov stability in the presence of a discretized plant. We begin by laying out the discrete linear approximation of the nonlinear equations

of motion, as presented in [7], where  $K_1, \dots, K_6$  are per-axis drag coefficients (see Table 4 for all coefficients) and  $\Omega_r = \omega_1 - \omega_2 + \omega_3 - \omega_4$  is the net rotor speed differential.

$$\begin{aligned} x_{k+2} = & 2x_{k+1} - x_k - \Delta t K_1 (x_{k+1} - x_k) / m \\ & + \Delta t^2 (\text{c } \phi_k \text{ s } \theta_k \text{ c } \psi_k + \text{s } \phi_k \text{ s } \psi_k) u_{1,k} / m \end{aligned} \quad (20)$$

$$\begin{aligned} y_{k+2} = & 2y_{k+1} - y_k - \Delta t K_2 (y_{k+1} - y_k) / m \\ & + \Delta t^2 (\text{c } \phi_k \text{ s } \theta_k \text{ s } \psi_k - \text{s } \phi_k \text{ c } \psi_k) u_{1,k} / m \end{aligned} \quad (21)$$

$$\begin{aligned} z_{k+2} = & 2z_{k+1} - z_k - \Delta t K_3 (z_{k+1} - z_k) / m \\ & + \Delta t^2 (\text{c } \phi_k \text{ c } \theta_k u_{1,k} / m - g) \end{aligned} \quad (22)$$

$$\begin{aligned} \phi_{k+2} = & 2\phi_{k+1} - \phi_k + (\theta_{k+1} - \theta_k) (\psi_{k+1} - \psi_k) \frac{(J_{yy} - J_{zz})}{J_{xx}} \\ & + \Delta t^2 u_{2,k} l / J_{xx} + \Delta t J_r (\theta_{k+1} - \theta_k) \Omega_r / J_{xx} - \Delta t K_4 l (\phi_{k+1} - \phi_k) / J_{xx} \end{aligned} \quad (23)$$

$$\begin{aligned} \theta_{k+2} = & 2\theta_{k+1} - \theta_k + (\psi_{k+1} - \psi_k) (\phi_{k+1} - \phi_k) \frac{(J_{zz} - J_{xx})}{J_{yy}} \\ & + \Delta t^2 u_{3,k} l / J_{yy} - \Delta t J_r (\phi_{k+1} - \phi_k) \Omega_r / J_{yy} - \Delta t K_5 l (\theta_{k+1} - \theta_k) / J_{yy} \end{aligned} \quad (24)$$

$$\begin{aligned} \psi_{k+2} = & 2\psi_{k+1} - \psi_k + (\phi_{k+1} - \phi_k) (\theta_{k+1} - \theta_k) \frac{(J_{xx} - J_{yy})}{J_{zz}} \\ & + \Delta t^2 u_{4,k} / J_{zz} - \Delta t K_6 (\psi_{k+1} - \psi_k) / J_{zz} \end{aligned} \quad (25)$$

Continuing with the problem setup from [7], we now define the sliding surface of the fully actuated subsystem ( $z$  and  $\psi$ ).

$$s_{z,k} = \alpha_z (z_k^d - z_k) + (\dot{z}_k^d - \dot{z}_k), \quad s_{\psi,k} = \alpha_\psi (\psi_k^d - \psi_k) + (\dot{\psi}_k^d - \dot{\psi}_k). \quad (26)$$

The coefficients  $\alpha_z$  and  $\alpha_\psi$  are positive constants; and we assume the desired point to be constant, iteration-to-iteration, and so  $\dot{z}_k^d = 0$  and  $\dot{\psi}_k^d = 0$ ; where we define  $\dot{z}_k = (z_{k+1} - z_k) / \Delta t$ ,  $\dot{\psi}_k = (\psi_{k+1} - \psi_k) / \Delta t$  as first-order approximations of the vehicle velocity. This assumption is strong, but it holds for all but finitely many time steps under waypoint commands. To maintain an exponential rate of convergence, we design control laws that satisfy the exponential reaching conditions.

$$s_{z,k+1} - s_{z,k} = (-\eta_z s_{z,k}) \Delta t, \quad s_{\psi,k+1} - s_{\psi,k} = (-\eta_\psi s_{\psi,k}) \Delta t \quad (27)$$

where  $\eta_z, \eta_\psi > 0$ . We now show that if  $\Delta_{k+1,k} s_i = (-\eta_i s_{i,k}) \Delta t$ , the Lyapunov stability requirements presented in [7] are satisfied.

$$\begin{aligned} V_{i,k} &= \frac{1}{2} s_{i,k}^2 > 0 \quad (\text{Lyapunov positivity, holds trivially}) \\ \Delta V_{i,k} &\triangleq V_{i,k+1} - V_{i,k} = \frac{1}{2} (s_{i,k+1}^2 - s_{i,k}^2) < 0 \quad (\text{discrete negative definiteness}) \\ &= \frac{1}{2} (s_{i,k+1} - s_{i,k}) (s_{i,k+1} + s_{i,k}) \\ &= -\frac{1}{2} \eta_i \Delta t |s_{i,k}| (2 - \eta_i \Delta t) |s_{i,k}| < 0 \quad (\text{for appropriate } \eta_i) \end{aligned} \quad (28)$$

Therefore, if our control laws satisfy the exponential reaching law, the discrete Lyapunov stability requirements hold, and our system exponentially converges to the desired point. This matters for the ballistic-deployment test case, which subjects the controller to large initial deviations.

From here, our work diverges significantly from [7]. We derive different control laws for  $z$  and  $\psi$ , as the originally proposed control laws do not satisfy the reaching conditions stated above. We assume that the quadrotor is symmetric in the rotor plane, so  $J_{xx} = J_{yy}$ , as for almost all commercial quadrotors. This inertial symmetry is load-bearing in the yaw channel: it removes the  $(J_{xx} - J_{yy}) \omega_x \omega_y$  gyroscopic term from Euler's equation, leaving  $\dot{\omega}_z = M_z / J_{zz}$  (Equation 2), on which the  $\psi$  sliding law of Equation 30 is built. The roll and pitch laws, the deficit-coupling matrix  $D_k$ , and the ISS bound retain  $J_{xx}$  and  $J_{yy}$  as distinct parameters; an asymmetric vehicle would reintroduce the yaw cross-term but leave the structure of the rest of the analysis unchanged.



$$\begin{aligned}
u_{1,k} &= \frac{m}{c \phi_k c \theta_k} \left( -\alpha_z \dot{z}_k + \frac{K_3}{m} \dot{z}_k + g + \eta_z s_{z,k} \right), \\
\Delta_{k+1,k} s_z &= (-\eta_z s_{z,k}) \Delta t, \\
&= \alpha_z (z_k^d - z_{k+1}) - \frac{1}{\Delta t} \left[ 2z_{k+1} - z_k + \Delta t^2 (c \phi_k c \theta_k \frac{u_{1,k}}{m} - g) \right. \\
&\quad \left. - \frac{\Delta t K_3}{m} (z_{k+1} - z_k) - z_{k+1} \right] - \alpha_z (z_k^d - z_k) - \frac{z_{k+1} - z_k}{\Delta t}, \\
&= \alpha_z (z_k - z_{k+1}) - \frac{1}{\Delta t} \left[ \Delta t^2 (c \phi_k c \theta_k \frac{u_{1,k}}{m} - g) - \frac{\Delta t K_3}{m} (z_{k+1} - z_k) \right], \\
&= \alpha_z (z_k - z_{k+1}) - \Delta t \left( -\alpha_z \dot{z}_k + \frac{K_3}{m} \dot{z}_k + \eta_z s_{z,k} \right) + \frac{K_3}{m} (z_{k+1} - z_k),
\end{aligned} \tag{29}$$

$$\begin{aligned}
&\vdots \\
\Delta_{k+1,k} s_z &= (-\eta_z s_{z,k}) \Delta t. \\
u_{4,k} &= J_{zz} \left( -\alpha_\psi \dot{\psi}_k + \frac{K_6}{J_{zz}} \dot{\psi}_k + \eta_\psi s_{\psi,k} \right), \\
&\vdots \\
s_{\psi,k+1} &= \alpha_\psi (\psi_k^d - \psi_{k+1}) - \left( \dot{\psi}_k + \Delta t (-\alpha_\psi \dot{\psi}_k + \frac{K_6}{J_{zz}} \dot{\psi}_k + \eta_\psi s_{\psi,k}) - \frac{\Delta t K_6}{J_{zz}} \dot{\psi}_k \right), \\
&\vdots \\
\Delta_{k+1,k} s_\psi &= -\Delta t \eta_\psi s_{\psi,k}.
\end{aligned} \tag{30}$$

In implementation, the fully-actuated sliding variables are saturated to a band  $s_{z,k}, s_{\psi,k} \in [-c, c]$  before entering the control laws, with  $c = 2$  for the baseline formulation and  $c = 1.578$  for the tuned saturated formulation (Table 4). Outside the band the reaching step becomes the constant-rate  $\Delta_{k+1,k} s_i = -c \eta_i \Delta t \operatorname{sgn}(s_{i,k})$ , which still satisfies the discrete existence condition  $|s_{i,k+1}| < |s_{i,k}|$  for  $\eta_i \Delta t < 2$ ; the exponential reaching law above is recovered once  $|s_{i,k}| \leq c$ . In the saturated controller of Section VI.C, the clip is applied to  $s$  before the auxiliary-state correction that forms  $\tilde{s}$ .

Contrary to this formulation, [7] proposed the following control laws for the fully-actuated dimensions of the quadrotor:

$$u_{1,k} = \frac{m}{c \phi_k c \theta_k} \left\{ \frac{-\alpha_z \dot{z}_k + g + K_3 \dot{z}_k}{m + \eta_z s_{z,k}} \right\} \tag{31}$$

$$u_{4,k} = J_{zz} \left\{ \frac{-\alpha_\psi \dot{\psi}_k + K_6 \dot{\psi}_k}{J_{zz} + \eta_\psi s_{\psi,k}} \right\} \tag{32}$$

We substitute the two-step extrapolated dynamics along with the proposed control laws above into  $s_{i,k+1} - s_{i,k}$  and simplify.

$$\begin{aligned}
\Delta_{k+1,k} s_z &= -\alpha_z \Delta t \dot{z}_k - \Delta t \left[ \frac{-\alpha_z \dot{z}_k + g + K_3 \dot{z}_k}{m + \eta_z s_{z,k}} - g \right] + \frac{K_3}{m} \Delta t \dot{z}_k \\
&= -\alpha_z \Delta t \dot{z}_k + \Delta t \frac{\alpha_z \dot{z}_k}{m + \eta_z s_{z,k}} - \Delta t \frac{g}{m + \eta_z s_{z,k}} \\
&\quad - \Delta t \frac{K_3 \dot{z}_k}{m + \eta_z s_{z,k}} + \frac{\Delta t K_3 \dot{z}_k}{m} + g \Delta t \\
&\neq (-\eta_z s_{z,k}) \Delta t
\end{aligned} \tag{33}$$

$$\begin{aligned}
\Delta_{k+1,k} s_\psi &= -\alpha_\psi \Delta t \dot{\psi}_k - \Delta t \left[ \frac{-\alpha_\psi \dot{\psi}_k + K_6 \dot{\psi}_k}{J_{zz} + \eta_\psi s_{\psi,k}} - \frac{K_6}{J_{zz}} \dot{\psi}_k \right] \\
&= \alpha_\psi \Delta t \dot{\psi}_k \left( \frac{1}{J_{zz} + \eta_\psi s_{\psi,k}} - 1 \right) + \Delta t K_6 \dot{\psi}_k \left( \frac{1}{J_{zz}} - \frac{1}{J_{zz} + \eta_\psi s_{\psi,k}} \right) \\
&\neq (-\eta_\psi s_{\psi,k}) \Delta t
\end{aligned} \tag{34}$$

Therefore, the control laws proposed in [7] violate the exponential reaching condition, and we proceed using the control laws of Equations 29 and 30. We now derive the control laws for pitch and roll. We enforce the same exponential reaching laws, which, as we proved in Equation 28, satisfy the conditions for Lyapunov stability. For brevity, we derive  $u_2$ , the roll control law, which has the same derivation form as  $u_3$ .

$$\begin{aligned}
s_{\phi,k} &\triangleq -a_1 \dot{y}_k + a_2 \left( y_k^d - y_k \right) - a_3 \dot{\phi}_k + a_4 \left( \phi_k^d - \phi_k \right) \\
T_{y,k} &\triangleq c \phi_k s \theta_k s \psi_k - s \phi_k c \psi_k, \quad T_{x,k} \triangleq c \phi_k s \theta_k c \psi_k + s \phi_k s \psi_k \\
g_{1,k} &\triangleq \frac{T_{y,k}}{m}, \quad g_{2,k} \triangleq \frac{T_{x,k}}{m}, \quad \Omega_r \triangleq \omega_1 - \omega_2 + \omega_3 - \omega_4 \\
f_{1,k} &\triangleq \frac{1}{J_{xx}} \left( \dot{\theta}_k \dot{\psi}_k (J_{yy} - J_{zz}) + J_r \dot{\theta}_k \Omega_r - K_4 l \dot{\phi}_k \right) \\
f_{2,k} &\triangleq \frac{1}{J_{yy}} \left( \dot{\psi}_k \dot{\phi}_k (J_{zz} - J_{xx}) - J_r \dot{\phi}_k \Omega_r - K_5 l \dot{\theta}_k \right)
\end{aligned} \tag{35}$$

$$\begin{aligned}
y_{k+2} - y_{k+1} &= \Delta t \dot{y}_k \left( 1 - \frac{\Delta t K_2}{m} \right) + \frac{\Delta t^2 T_{y,k}}{m} u_{1,k} \implies \dot{y}_{k+1} - \dot{y}_k = \Delta t \left[ g_{1,k} u_{1,k} - \frac{K_2}{m} \dot{y}_k \right] \\
\phi_{k+2} - \phi_{k+1} &= \Delta t \dot{\phi}_k + \Delta t^2 \left[ f_{1,k} + \frac{u_{2,k} l}{J_{xx}} \right] \implies \dot{\phi}_{k+1} - \dot{\phi}_k = \Delta t \left[ f_{1,k} + \frac{u_{2,k} l}{J_{xx}} \right] \\
x_{k+2} - x_{k+1} &= \Delta t \dot{x}_k \left( 1 - \frac{\Delta t K_1}{m} \right) + \frac{\Delta t^2 T_{x,k}}{m} u_{1,k} \implies \dot{x}_{k+1} - \dot{x}_k = \Delta t \left[ g_{2,k} u_{1,k} - \frac{K_1}{m} \dot{x}_k \right] \\
\theta_{k+2} - \theta_{k+1} &= \Delta t \dot{\theta}_k + \Delta t^2 \left[ f_{2,k} + \frac{u_{3,k} l}{J_{yy}} \right] \implies \dot{\theta}_{k+1} - \dot{\theta}_k = \Delta t \left[ f_{2,k} + \frac{u_{3,k} l}{J_{yy}} \right]
\end{aligned} \tag{36}$$

$$\delta y_k \triangleq g_{1,k} u_{1,k} - \frac{K_2}{m} \dot{y}_k, \quad \delta x_k \triangleq g_{2,k} u_{1,k} - \frac{K_1}{m} \dot{x}_k \tag{37}$$

Roll-axis surface:

$$\begin{aligned}
\Delta_{k+1,k} s_{\phi} &= \Delta t \left[ -a_1 \delta y_k - a_2 \dot{y}_k - a_3 \left( f_{1,k} + \frac{u_{2,k} l}{J_{xx}} \right) - a_4 \dot{\phi}_k \right] = -\eta_{\phi} s_{\phi,k} \Delta t, \\
u_{2,k} &= \frac{J_{xx}}{l a_3} \left[ -a_1 \delta y_k - a_2 \dot{y}_k - a_3 f_{1,k} - a_4 \dot{\phi}_k + \eta_{\phi} s_{\phi,k} \right].
\end{aligned} \tag{38}$$

Pitch-axis surface:

$$\begin{aligned}
\Delta_{k+1,k} s_{\theta} &= \Delta t \left[ -a_5 \delta x_k - a_6 \dot{x}_k - a_7 \left( f_{2,k} + \frac{u_{3,k} l}{J_{yy}} \right) - a_8 \dot{\theta}_k \right] = -\eta_{\theta} s_{\theta,k} \Delta t, \\
u_{3,k} &= \frac{J_{yy}}{l a_7} \left[ -a_5 \delta x_k - a_6 \dot{x}_k - a_7 f_{2,k} - a_8 \dot{\theta}_k + \eta_{\theta} s_{\theta,k} \right].
\end{aligned} \tag{39}$$

## IV. Simulation Design

### A. Ballistic Trajectory

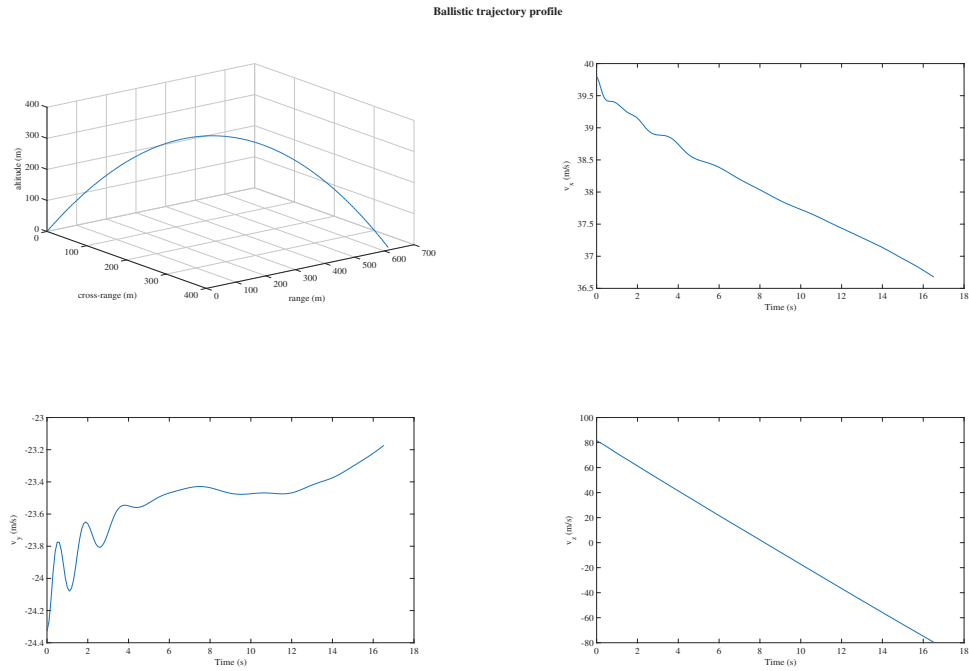
We simulate a six degrees of freedom (6-DOF) ballistic trajectory using the equations outlined in Chapter 9 of [17]. For the moment, we use the standard model of a 120 mm mortar shell given in [17], due to readily available aerodynamic characteristics and our focus on the family of initial conditions generated by the ballistic trajectory, rather than the particular trajectory. The specific trajectory parameters are given in Table 6. The launch elevation is chosen to place the deployment envelope within the controller's recoverable regime: a shallow (45°) launch produces a fast ( $\approx 70$  m/s) apogee deploy that exceeds the arrestable envelope, whereas the steeper 64° launch trades range for a gentler ( $\approx 45$  m/s) deploy. The azimuth in Table 6 is McCoy's departure angle from the range plane, not a compass heading.

**Table 4** Gains and scalar parameters. Reaching-law gains, sliding-surface coefficients and the sliding-variable clip are those of the saturated dSMC, set by a global Latin-hypercube search over landing reachability; the parenthesized value is the untuned baseline set that the no-saturation formulation still runs.

| Parameter          | Value               | Parameter         | Value                         |
|--------------------|---------------------|-------------------|-------------------------------|
| $K_1$              | 0.0001              | $K_2$             | 0.0001                        |
| $K_3$              | 0.0001              | $K_4$             | 0.0012                        |
| $K_5$              | 0.0012              | $K_6$             | 0.0012                        |
| $\eta_z$           | 7.3044 (7)          | $\eta_\psi$       | 8.2874 (7)                    |
| $\eta_\phi$        | 4.9968 (14)         | $\eta_\theta$     | 4.9968 (14)                   |
| $\alpha_z$         | 10.4815 (6)         | $\alpha_\psi$     | 3.1535 (1)                    |
| $b$                | 5                   | $d$               | 2                             |
| $\Omega_{\min}^2$  | 0                   | $\Omega_{\max}^2$ | 2                             |
| $s_z, s_\psi$ clip | $\pm 1.578 (\pm 2)$ | $k_{\xi, \alpha}$ | $1.2741 \eta_\alpha \Delta t$ |

**Table 5** Coefficient definitions  $a_i$ , for the saturated dSMC. The untuned baseline formulation uses 6, 2, 2 and 8 in place of the four leading constants, mirrored in  $a_5$ – $a_8$ .

| $a_i$ | Definition                            | $a_i$ | Definition                            |
|-------|---------------------------------------|-------|---------------------------------------|
| $a_1$ | $\frac{5.1762 m}{u_T c \psi}$         | $a_2$ | $\frac{0.7410 m}{u_T c \psi}$         |
| $a_3$ | 3.4330                                | $a_4$ | 14.6286                               |
| $a_5$ | $-\frac{5.1762 m}{u_T c \phi c \psi}$ | $a_6$ | $-\frac{0.7410 m}{u_T c \phi c \psi}$ |
| $a_7$ | 3.4330                                | $a_8$ | 14.6286                               |



**Fig. 2** Mortar shell ballistic trajectory. States sampled along this trajectory are the initial conditions for the drone deployment simulations.

**Table 6** Launch parameters for the 120 mm mortar ballistic trajectory used as the deployment-point envelope for the dSMC reachability sweep.

| Symbol                                                 | Description                                  | Value  | Unit  |
|--------------------------------------------------------|----------------------------------------------|--------|-------|
| <b>Launch kinematics</b>                               |                                              |        |       |
| $V_0$                                                  | Muzzle velocity                              | 94     | m/s   |
| $\theta_e$                                             | Launch elevation angle                       | 64     | deg   |
| $\theta_a$                                             | Launch azimuth angle                         | 15     | deg   |
| <b>Initial angular state</b>                           |                                              |        |       |
| $p_0$                                                  | Initial axial (roll) spin rate               | -8.379 | rad/s |
| $\omega_{y,0}$                                         | Initial transverse angular rate (yaw axis)   | 0.5    | rad/s |
| $\omega_{z,0}$                                         | Initial transverse angular rate (pitch axis) | 1      | rad/s |
| $\alpha_0$                                             | Initial angle of attack (pitch misalignment) | 2      | deg   |
| $\beta_0$                                              | Initial sideslip angle (yaw misalignment)    | -0.5   | deg   |
| <b>Initial position (range, altitude, cross-range)</b> |                                              |        |       |
| $x_0$                                                  | Range-axis launch position                   | 0      | m     |
| $y_0$                                                  | Altitude-axis launch position                | 0      | m     |
| $z_0$                                                  | Cross-range-axis launch position             | 0      | m     |
| <b>Integrator</b>                                      |                                              |        |       |
| $t_{\max}$                                             | Maximum propagation time                     | 300    | s     |

## B. Parameters and Configuration

We simulate our system in MATLAB Simulink R2025a. The system is discretized using 50 Hz zero-order hold (ZOH) blocks placed before and after the controller plant. The dynamics are continuously integrated, but sampled using ZOHs. The closed-loop drone simulations use ‘auto’ solver selection with default integration settings; the 6-DOF ballistic propagation instead uses tightened tolerances (RelTol  $10^{-8}$ , AbsTol  $10^{-10}$ ) on a fixed 0.1 s output grid. The Simulink models and analysis scripts are available at [18].

# V. Simulation Results

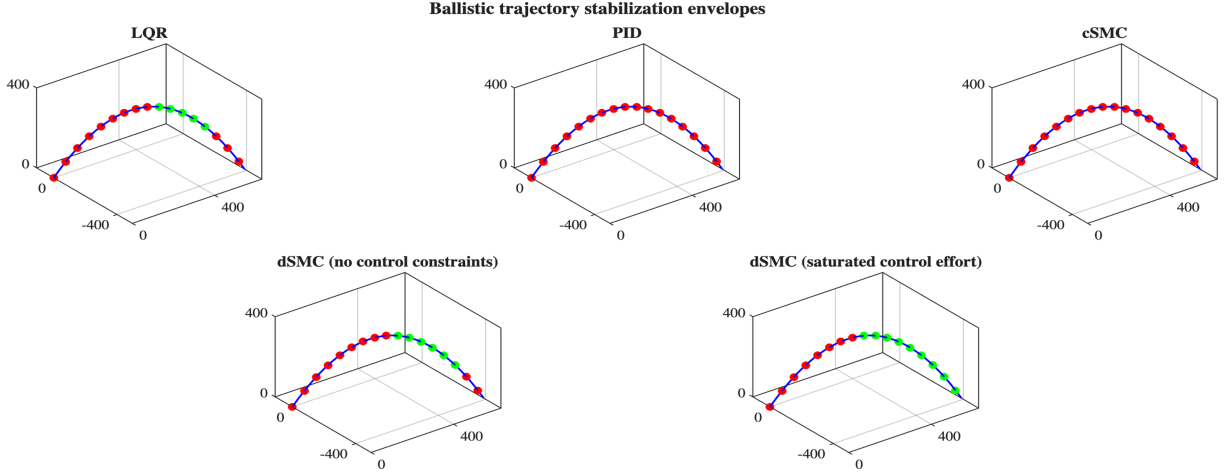
## A. Ballistic Flight Envelope

At evenly spaced points along the ballistic trajectory (Figure 2), we sampled the vehicle state and ran a 30-second stabilization attempt from each sample, recording which initial conditions each controller could recover from. Each shell state maps to the drone’s 12-state initial condition by aligning the body  $x$ -axis with the pointing vector (zero roll) and rotating the shell’s linear and angular velocity into that body frame. A trial was counted as successful if the vehicle survived the full stabilization window without the divergence guard firing and satisfied the three conditions below:

$$\sqrt{(x_t - x_f)^2 + (y_t - y_f)^2} \leq 10 \text{ m}, \quad \frac{|v_t|_2}{|v_0|_2} \leq 2, \quad \frac{|\dot{\Theta}_t|_2}{|[\pi \pi \pi]^\top|_2} \leq 2 \quad (t > 1 \text{ s}). \quad (40)$$

The first condition bounds the final horizontal position error to within 10 m of the target. The second prevents the controller from accelerating the vehicle past twice its initial deployment speed  $|v_0|_2$ . The third caps the Euler-angle rate  $\dot{\Theta}_t = [\dot{\theta} \dot{\phi} \dot{\psi}]$  at twice  $|[\pi \pi \pi]^\top|_2 = \pi\sqrt{3} \approx 5.44 \text{ rad/s}$ , the magnitude of a rate vector with  $\pi \text{ rad/s}$  on each axis. The rotational-rate bound is evaluated only after the first second of flight: the vehicle is released tumbling, so rate excursions inside that window reflect the ballistic-separation transient the controller is still arresting, not the controller itself. Of these conditions only the velocity bound can be re-scored from stored trial data without re-simulating, since it enters as the single inequality  $|v_t|_2/|v_0|_2 \leq V$  evaluated on a logged trace; re-scoring every dSMC trial at  $V \in \{2, 3, 5, 10, \infty\}$  leaves all four of the counts we report unchanged—6 and 9 of 17 on this envelope, and 31 and 85 of 140 on the landing sweep of Section VI.I. The largest speed ratio reached by any trial that satisfies the other conditions is 1.40, so no trial sits near the threshold and the reported separations do not depend on where it is placed. That insensitivity is in part a property of the plant: the simulated vehicle integrates rigid-body dynamics under gravity and thrust with no aerodynamic drag, so no terminal velocity arises on its own, failures diverge instead of settling at some large speed, and this condition stands in for the speed limit a real airframe would impose. The drag coefficients retained in the implemented thrust and yaw laws ( $K_3$  and  $K_6$  of Table 4) accordingly compensate a term the simulated plant does not contain; at these values that correction is orders of magnitude below the gravitational term and does not affect the results reported here. The criterion is scored on horizontal position, whereas our stability analysis certifies the four

actuated sliding variables (altitude, roll, pitch, and yaw); horizontal  $(x, y)$  motion rides on the underactuated attitude channels, and its convergence is shown empirically here, not proven. Figure 3 shows that no controller recovers every sampled state; the dSMC recovers the largest share (the saturated variant, 9 of 17 sampled states), concentrated on the descending leg past apogee. LQR and PID operate too far from their hover linearization, and cSMC’s continuous-time stability argument does not transfer to the sampled-data implementation. Each baseline uses gains tuned for stable hover and step tracking (Tables 2 and 3, and the LQR weights of Equation 5), not detuned to favor the dSMC; the failures here are structural—a linearization or continuity assumption broken by the tumbling deploy—rather than an artifact of gain choice. We did not exhaustively search each baseline’s gain space, so we report the separation as observed on this envelope rather than as a gain-independent ranking.



**Fig. 3 Ballistic stabilization at test points along the trajectory. Recovered states per controller (of 17 sampled): LQR 5, PID 0, cSMC 0, dSMC without saturation 6, dSMC with priority-weighted saturation 9. The two dSMC formulations differ in gain set as well as saturation handling (Table 4), so the 6-to-9 margin is not attributable to the saturation handler alone. Isolating the handler at matched tuned gains and the same  $\Omega_{\max}^2 = 2$ , simple per-motor clipping recovers 3 of 17 and the priority-weighted allocation 9 of 17; the allocation gives up no state that clipping recovers.**

## VI. Input Saturation Handling for the Discrete Sliding Mode Controller

The dSMC above produces an unconstrained command  $u_k = (T_k, M_{x,k}, M_{y,k}, M_{z,k})^T \in \mathbb{R}^4$ . Real motors enforce per-rotor speed limits, whether due to electrical limitations of the motor or its electronic speed controller, or aerostructural limitations of the rotor itself. These limitations define a feasible set  $\mathcal{U} \subset \mathbb{R}^4$  for the applied command signals, namely, per-motor rotation speeds. When  $u_k \notin \mathcal{U}$ , the controller has to project onto  $\mathcal{U}$  and account for the resulting tracking deficit. Inside  $\mathcal{U}$ , and provided no saturation has yet occurred, the closed loop is Lyapunov stable, as the controller coincides with the unconstrained dSMC; once a saturation episode ends, the controller instead converges back to the baseline dSMC as the anti-windup state decays (Section VI.G). Outside  $\mathcal{U}$ , it relaxes to an ISS bound, which we will show below.

### A. Motor Mixing and Saturation Limits

Rotor-speed-squared  $\Omega^2 = (\Omega_1^2, \Omega_2^2, \Omega_3^2, \Omega_4^2)^T$  relates to the command via the symmetric-quadrotor mixing matrix

$$u = T_M \Omega^2, \quad T_M = \begin{bmatrix} b & b & b & b \\ 0 & -b & 0 & b \\ -b & 0 & b & 0 \\ d & -d & d & -d \end{bmatrix}, \quad (41)$$

where  $b$  is the per-rotor thrust-per- $\Omega^2$  coefficient,  $d$  the analogous yaw-torque coefficient, and the roll/pitch moment rows carry an effective moment coefficient that we set numerically equal to  $b$ . We work in normalized  $\Omega^2$  units with the implementation values  $b = 5$ ,  $d = 2$ ,  $\Omega_{\min}^2 = 0$ , and  $\Omega_{\max}^2 = 2$  (Table 4); these describe a notional actuator with thrust-to-weight ratio  $4b\Omega_{\max}^2/(mg) = 5.10$  and hover point  $\Omega_{\text{hover}}^2 = mg/(4b) = 0.392$  strictly inside the feasible band, and are effective values for the simulated actuator rather than quantities derived from the physical mixing of Equation 1. Extending the component-wise discrete-time saturation framework of Wang et al. [13] to the per-rotor case (which we choose due to the underlying source of input constraints being motor speed limits, and so is more physically connected),  $\Omega_i^2 \in [\Omega_{\min}^2, \Omega_{\max}^2]$  defines the feasible set in command space as  $\mathcal{U} = T_M \cdot [\Omega_{\min}^2, \Omega_{\max}^2]^4$ . Let  $\bar{u}_k \in \mathcal{U}$  denote the actually applied (after saturation) command and define the saturation deficit

$$\Delta u_k \triangleq u_k - \bar{u}_k. \quad (42)$$

The rows of  $T_M$  are pairwise orthogonal by inspection, so the Gram matrix is diagonal:

$$T_M T_M^T = \text{diag}(4b^2, 2b^2, 2b^2, 4d^2), \quad (T_M T_M^T)^{-1} = \text{diag}\left(\frac{1}{4b^2}, \frac{1}{2b^2}, \frac{1}{2b^2}, \frac{1}{4d^2}\right). \quad (43)$$

Now, we can scale channel-by-channel in the closed-form projection below. Scaling the command on a single channel scales one column-contribution of  $T_M^{-1} u$  in  $\Omega^2$ -space—one column of  $T_M^{-1}$  times the corresponding component of  $u$ , the  $v_\alpha$  of Equation 46—and leaves the other three contributions untouched. We adopt the ranking presented in Faessler et al. [11], which ranks roll and pitch above total thrust above yaw; the low priority of yaw is consistent with Mueller and D’Andrea’s [12] treatment of yaw rotation as dynamically inessential for trajectory interception. In simulation, we observed that when the rotor limit passes the priority-allocation feasibility boundary, roll and pitch divergence lead to destabilization and crashes well before yaw or altitude tracking degrades. We therefore adopt the priority order

$$M_x, M_y \succ T \succ M_z, \quad (44)$$

and enforce the control ordering by successively saturating the control channels. As an example, if a control effort can be made feasible by strictly limiting yaw torque, we will use it instead of an alternate strategy which saturates both yaw and collective thrust. Indexing the four control channels as  $\alpha \in \{T, M_x, M_y, M_z\}$ , the deficit vector  $\Delta u_k$ , auxiliary state  $\xi_k$ , and gain matrix  $E = \Delta t \text{diag}(\eta_T, \eta_{M_x}, \eta_{M_y}, \eta_{M_z})$  all follow the  $\{T, M_x, M_y, M_z\}$  ordering, with the channel gains identified with the dSMC gains of Table 4 through  $(\eta_T, \eta_{M_x}, \eta_{M_y}, \eta_{M_z}) \equiv (\eta_z, \eta_\phi, \eta_\theta, \eta_\psi)$ . The numeric bounds of Section VI.F and the residual ratios of Section VI.H both use this identification.

## B. Priority-Weighted Allocation

The projection realizing the priority of Equation 44 acts directly on the per-channel contributions in motor space and admits a closed form. Where Faessler et al. [11] achieve the same hierarchy through an iterative mixer with no stability guarantee, the construction here runs in fixed time per tick and admits the Lyapunov stability proof of Section VI.F.

Using  $T_M^{-1} = T_M^T (T_M T_M^T)^{-1}$  together with Equation 43,

$$T_M^{-1} = \begin{bmatrix} b & 0 & -b & d \\ b & -b & 0 & -d \\ b & 0 & b & d \\ b & b & 0 & -d \end{bmatrix} \text{diag}\left(\frac{1}{4b^2}, \frac{1}{2b^2}, \frac{1}{2b^2}, \frac{1}{4d^2}\right) = \begin{bmatrix} \frac{1}{4b} & 0 & -\frac{1}{2b} & \frac{1}{4d} \\ \frac{1}{4b} & -\frac{1}{2b} & 0 & -\frac{1}{4d} \\ \frac{1}{4b} & 0 & \frac{1}{2b} & \frac{1}{4d} \\ \frac{1}{4b} & \frac{1}{2b} & 0 & -\frac{1}{4d} \end{bmatrix}. \quad (45)$$

Applying  $T_M^{-1}$  to a command  $u = (T, M_x, M_y, M_z)^T$  and collecting columns,

$$\Omega^2 = T_M^{-1} u = \underbrace{\frac{T}{4b} \mathbf{1}}_{v_T} + \underbrace{\frac{M_x}{2b} e_{M_x}}_{v_{M_x}} + \underbrace{\frac{M_y}{2b} e_{M_y}}_{v_{M_y}} + \underbrace{\frac{M_z}{4d} e_{M_z}}_{v_{M_z}}, \quad (46)$$

where  $\mathbf{1} = (1, 1, 1, 1)^T$ ,  $e_{M_x} = (0, -1, 0, 1)^T$ ,  $e_{M_y} = (-1, 0, 1, 0)^T$ ,  $e_{M_z} = (1, -1, 1, -1)^T$  are the sign patterns of the rows of  $T_M$ . The vectors  $v_\alpha \in \mathbb{R}^4$  are the per-channel contributions to motor-speed-squared, distinct from the deficit components  $c_\alpha$  used in Section VI.H. The four basis vectors  $\{\mathbf{1}, e_{M_x}, e_{M_y}, e_{M_z}\}$  are mutually orthogonal in  $\mathbb{R}^4$  (they are columns of  $T_M^T$ , whose Gram matrix is diagonal). Separately, the additive structure of Equation 46 maps each command channel to a single contribution  $v_\alpha$ , so replacing the command on channel  $\alpha$  by  $\lambda u_\alpha$  scales only  $v_\alpha$  by  $\lambda$  and

leaves the other three contributions untouched. Each component of  $\Omega^2(\lambda)$  is then an affine function of the single scalar  $\lambda$ , and componentwise feasibility  $\Omega_i^2(\lambda) \in [\Omega_{\min}^2, \Omega_{\max}^2]$  reduces to four scalar interval constraints, solved in closed form by

$$\lambda_{\max}(\text{base}, \text{dir}) = \max\left(0, \min\left(1, \min_{i: \text{dir}_i \neq 0} \left\{ \begin{array}{ll} \frac{\Omega_{\max}^2 - \text{base}_i}{\text{dir}_i} & \text{if } \text{dir}_i > 0 \\ \frac{\Omega_{\min}^2 - \text{base}_i}{\text{dir}_i} & \text{if } \text{dir}_i < 0 \end{array} \right\}\right)\right), \quad (47)$$

returning 0 if any component with  $\text{dir}_i = 0$  has  $\text{base}_i \notin [\Omega_{\min}^2, \Omega_{\max}^2]$ . The implementation additionally tracks the lower interval endpoints contributed by the opposite box face and returns 0 when the per-component intervals for the scale have empty intersection, so an unreachable stage yields a zero scale rather than an unguaranteed one. When the base point is itself feasible, the two forms coincide and the point returned by Equation 47 is feasible; when it is not, the returned scale carries no such guarantee. The trailing feasibility checks in the cascade below are therefore load-bearing rather than redundant: each stage short of the terminal fallback returns its candidate only after re-checking the full motor-space point.

We apply cascaded saturation by successively saturating channels, in ascending order of importance, until we reach a feasible control effort. So, we accept relatively minor errors in exchange for controlling more important channels.

- 1) The squared rotor speed  $\Omega^2 = T_M^{-1} u_k$ . If  $\Omega^2 \in [\Omega_{\min}^2, \Omega_{\max}^2]^4$ , then our control effort satisfies our control limits, and we simply return  $\bar{u}_k = u_k$  and  $\Delta u_k = 0$ , which is the unmodified dSMC control effort.
- 2) Otherwise, our control effort is infeasible, and we begin the saturation process. We start by scaling the lowest-priority channel  $M_z$  by  $\alpha_{M_z} \in [0, 1]$  while leaving  $T$ ,  $M_x$ ,  $M_y$  at their commanded values:

$$\Omega^2(\alpha_{M_z}) = \underbrace{v_T + v_{M_x} + v_{M_y}}_{\text{base}} + \alpha_{M_z} \underbrace{v_{M_z}}_{\text{dir}}. \quad (48)$$

With  $\alpha_{M_z} = \lambda_{\max}(\text{base}, v_{M_z})$ , return  $\bar{u}_k = (T, M_x, M_y, \alpha_{M_z} M_z)^T$  if feasible. In this regime, we begin to sacrifice some yaw control, but retain altitude and attitude control.

- 3) If yaw scaling is unable to produce a feasible control effort, we temporarily abandon yaw control and begin scaling collective thrust by  $\beta_T \in [0, 1]$ , which affects our altitude control:

$$\Omega^2(\beta_T) = \underbrace{v_{M_x} + v_{M_y} + \beta_T}_{\text{base}} \underbrace{v_T}_{\text{dir}}. \quad (49)$$

With  $\beta_T = \lambda_{\max}(v_{M_x} + v_{M_y}, v_T)$ , return  $\bar{u}_k = (\beta_T T, M_x, M_y, 0)^T$  if the control effort is feasible. In this case, we maintain full roll and pitch authority, limited altitude authority, and completely discard our yaw authority.

- 4) If thrust scaling is infeasible, then zero collective thrust and zero yaw cannot accommodate the demanded  $(M_x, M_y)$ . In this case, we scale the joint roll/pitch contribution by  $\gamma_{M_{xy}} = \lambda_{\max}(0, v_{M_x} + v_{M_y}) \in [0, 1]$  and return  $\bar{u}_k = (0, \gamma_{M_{xy}} M_x, \gamma_{M_{xy}} M_y, 0)^T$ . This terminal stage assumes  $\Omega_{\min}^2 = 0$ ; with any strictly positive idle floor, no point on the zero-thrust ray is feasible and the closed form admits no feasible scale. Even under that assumption, the closed form is blunt: the components of the direction  $v_{M_x} + v_{M_y} = \frac{1}{2b} (-M_y, -M_x, M_y, M_x)^T$  occur in  $\pm$  pairs, so at least one is strictly negative whenever  $(M_x, M_y) \neq (0, 0)$ , and with a zero base Equation 47 returns  $\gamma_{M_{xy}} = 0$ . The terminal fallback is therefore  $\bar{u}_k = 0$ , a full motor cutoff that retains no control authority on any channel for that tick, rather than a scaled roll/pitch command. The severity of this stage should intuitively indicate why all stability proofs of our saturated controller require saturation to be transient—upon reaching stage 4 the vehicle coasts unactuated, and recovery is possible only once the commanded moments shrink enough for an earlier stage of the cascade to return a feasible effort.

### C. Auxiliary State and Modified Sliding Variable

The projection  $u_k \mapsto \bar{u}_k$  generates  $\Delta u_k \neq 0$  whenever saturation is required. Were we to simply substitute  $u_k \mapsto \bar{u}_k$  into our previous derivation of the sliding surface, the control deficit would violate the reaching condition  $\Delta s_k = -\eta_\alpha \Delta t s_k$ , and invalidate our stability proofs. Therefore, we need to modify the sliding surface using auxiliary states in order to account for this control deficit. We follow the basic concepts of the discrete-time anti-windup compensator of Wang et al. [13], in which an auxiliary state is driven by the saturation deficit  $\sigma(u) - u$  to recover stability under input saturation. Adapting the continuous-time sliding-mode auxiliary-state constructions of [8, 9] into discrete time across all four actuator channels, we introduce, for each  $\alpha \in \{T, M_x, M_y, M_z\}$ , a scalar state  $\xi_\alpha$  with update

$$\xi_{\alpha k+1} = (1 - k_{\xi \alpha}) \xi_{\alpha k} + \Delta u_{\alpha k}, \quad 0 < k_{\xi \alpha} < 2, \quad (50)$$

i.e., a discrete-time first-order filter driven by the per-channel deficit, pole at  $1 - k_{\xi \alpha}$ . In matrix form,  $\xi_{k+1} = (I - K_{\xi}) \xi_k + \Delta u_k$  with  $K_{\xi} = \text{diag}(k_{\xi \alpha})$ . The diagonal structure obtained by using one scalar state per actuator channel (rather than the vector-valued second-order auxiliary subsystem in [9] or the attitude-only per-channel construction in [8]) is what enables the diagonal Lyapunov decomposition in Section VI.F. Defining

$$\tilde{s}_k \triangleq s_k - D_k \xi_k, \quad (51)$$

where  $D_k \in \mathbb{R}^{4 \times 4}$  is the deficit-coupling matrix derived in the following subsection, the baseline dSMC control laws are reused verbatim with  $s_{\alpha} \mapsto \tilde{s}_{\alpha}$  substituted in the reaching-law terms of Equations 29–30 and their underactuated analogs; the equivalent-control terms are unchanged. So,  $D_k \xi_k$  is the cumulative effect of past deficits on the sliding manifold, and subtracting it from  $s_k$  gives the controller an estimate of what the sliding variable would have been without saturation. The cancellation occurs in Equation 76.

Each control channel of the auxiliary state has its own contraction variable,  $1 - k_{\xi \alpha}$ . We choose

$$k_{\xi \alpha} = \eta_{\alpha} \Delta t, \quad (52)$$

which is the same rate at which the unconstrained reaching law drives  $s$  to zero. This way, we can reuse the dSMC gains directly via Equation 52, as in the zero-control-deficit case, we recover the unsaturated dSMC controller. Both blocks of the closed-loop matrix  $\mathcal{A}_k$  (Equation 75) then decay at consistent per-channel rates, and the steady-state residual analysis of Section VI.H becomes  $\Delta t$ -independent ratios. The implementation scales all four channels uniformly,  $k_{\xi \alpha} = 1.2741 \eta_{\alpha} \Delta t$  (Table 4), the scale factor having been carried through the same tuning search as the gains. The certificates below do not require the matched choice: they are stated for any diagonal  $K_{\xi}$  satisfying Equation 50, the one bound in which  $K_{\xi}$  and  $E$  separate is evaluated at the implemented scaling (Equation 89), and the residual ratios of Section VI.H are independent of  $K_{\xi}$  altogether. When the commanded per-motor speeds lie within the feasible range,  $\Delta u_k = 0$  forces  $\xi$  to contract to zero, and the modified sliding variable reduces to the baseline  $s_k$  (Section VI.G).

#### D. Deficit-Coupling Matrix

We now derive the coupling matrix  $D_k$  between the auxiliary state and the control deficit. Two related coefficients enter the analysis and must be distinguished. Define

$$A_{ij} \triangleq \frac{\partial(\Delta s_i)}{\partial u_j} \quad (\text{formula coefficient: how commanded } u_j \text{ enters } \Delta s_i) \quad (53)$$

$$C_{ij} \triangleq \frac{\partial(\Delta s_i)}{\partial(\Delta u_j)} \quad (\text{deficit-response coefficient: how the deficit propagates}). \quad (54)$$

Decompose the intended surface change as

$$\Delta s_i|_{\text{commanded}} = A_{ij} u_j + (\text{terms independent of } u_j). \quad (55)$$

The saturated input is  $\bar{u}_j = u_j - \Delta u_j$ , and so the surface change corresponding to this saturated input is

$$\Delta s_i|_{\text{actual}} = A_{ij} \bar{u}_j + (\text{rest}) = \Delta s_i|_{\text{commanded}} - A_{ij} \Delta u_j. \quad (56)$$

The  $\partial/\partial(\Delta u_j)$  on the right-hand side gives

$$C = -A. \quad (57)$$

The matrix used in the substitution Equation 51 must equal  $C$  for the deficit to cancel in the  $\tilde{s}$  dynamics (verified in Equation 76 below), so

$$D = C = -A. \quad (58)$$

The six nonzero entries follow from direct differentiation of the discrete sliding-variable increments, holding the coefficients  $a_i$  over the step, consistent with the baseline dSMC derivation; their step-to-step variation is part of the  $D_{k+1} \approx D_k$  remainder of Section VI.E. For the fully-actuated thrust  $\rightarrow s_z$  entry  $D_{1,1}$ , the two-step extrapolated altitude dynamics of [7] give the velocity increment

$$\dot{z}_{k+1} - \dot{z}_k = \Delta t \left[ \frac{c \phi_k c \theta_k}{m} u_{1,k} - g - \frac{K_3}{m} \dot{z}_k \right]. \quad (59)$$



With  $s_{z,k} = \alpha_z(z_k^d - z_k) + (\dot{z}_k^d - \dot{z}_k)$  and constant reference state inputs, the increment is:

$$\Delta_{k+1,k} s_z = -\alpha_z \Delta t \dot{z}_k - \Delta t \left[ \frac{c \phi_k c \theta_k}{m} u_{1,k} - g - \frac{K_3}{m} \dot{z}_k \right]. \quad (60)$$

Only the  $-\Delta t c \phi_k c \theta_k u_{1,k}/m$  term contains  $u_{1,k}$ , giving

$$A_{1,1} = -\frac{\Delta t c \phi_k c \theta_k}{m}, \quad D_{1,1} = \frac{\Delta t c \phi_k c \theta_k}{m}. \quad (61)$$

For the fully-actuated yaw  $\rightarrow s_\psi$  entry  $D_{4,4}$ , the two-step extrapolated yaw dynamics give  $\dot{\psi}_{k+1} - \dot{\psi}_k = \Delta t [u_{4,k}/J_{zz} - (K_6/J_{zz}) \dot{\psi}_k]$ . The same procedure applied to  $s_{\psi,k} = \alpha_\psi(\psi_k^d - \psi_k) + (\dot{\psi}_k^d - \dot{\psi}_k)$  yields

$$A_{4,4} = -\frac{\Delta t}{J_{zz}}, \quad D_{4,4} = \frac{\Delta t}{J_{zz}}. \quad (62)$$

The underactuated roll entries  $D_{2,1}, D_{2,2}$  follow from

$$s_{\phi,k} = -a_1 \dot{y}_k + a_2(y_k^d - y_k) - a_3 \dot{\phi}_k + a_4(\phi_k^d - \phi_k). \quad (63)$$

With a constant reference state, the surface evolution expands as

$$\Delta_{k+1,k} s_\phi = -a_1 (\dot{y}_{k+1} - \dot{y}_k) - a_2 (y_{k+1} - y_k) - a_3 (\dot{\phi}_{k+1} - \dot{\phi}_k) - a_4 (\phi_{k+1} - \phi_k). \quad (64)$$

Substituting the two-step linearly extrapolated dynamics,

$$\dot{y}_{k+1} - \dot{y}_k = \Delta t \left[ g_{1,k} u_{1,k} - \frac{K_2}{m} \dot{y}_k \right] =: \Delta t \delta y_k, \quad (65)$$

$$\dot{\phi}_{k+1} - \dot{\phi}_k = \Delta t \left[ f_{1,k} + \frac{u_{2,k} l}{J_{xx}} \right], \quad (66)$$

$$y_{k+1} - y_k = \Delta t \dot{y}_k, \quad \phi_{k+1} - \phi_k = \Delta t \dot{\phi}_k, \quad (67)$$

gives

$$\Delta_{k+1,k} s_\phi = \Delta t \left[ -a_1 \delta y_k - a_2 \dot{y}_k - a_3 \left( f_{1,k} + \frac{u_{2,k} l}{J_{xx}} \right) - a_4 \dot{\phi}_k \right]. \quad (68)$$

Two control-dependent terms appear:  $\delta y_k = g_{1,k} u_{1,k} - K_2 \dot{y}_k/m$  is dependent on  $u_{1,k}$ , thrust, and  $a_3 u_{2,k} l/J_{xx}$  is dependent on  $u_{2,k}$ , the roll moment. Differentiating,

$$\Delta_{k+1,k} s_\phi|_u = -\Delta t a_1 g_{1,k} u_{1,k} - \frac{\Delta t a_3 l}{J_{xx}} u_{2,k}, \quad (69)$$

yielding

$$A_{2,1} = -\Delta t a_1 g_{1,k}, \quad A_{2,2} = -\frac{\Delta t a_3 l}{J_{xx}}, \quad (70)$$

and, by  $D = -A$ ,  $D_{2,1} = \Delta t a_1 g_{1,k}$ ,  $D_{2,2} = \Delta t a_3 l/J_{xx}$ . One implementation point matters here: the  $\delta y_k$  term inside the wrapped roll law (and its pitch analog  $\delta x_k$ ) must be evaluated with the commanded  $u_{1,k}$ , not the saturated  $\bar{u}_{1,k}$ , since Equation 69 differentiates with respect to the commanded input. Evaluating it with  $\bar{u}_{1,k}$  changes the effective  $D_{2,1}$  and  $D_{3,1}$  and leaves an unmodeled residual in the cancellation of Section VI.E. The pitch entries  $D_{3,1}, D_{3,3}$  are structurally the same, but replacing  $\{a_1, a_2, a_3, a_4\}$  with  $\{a_5, a_6, a_7, a_8\}$  respectively,  $g_{1,k}$  with  $g_{2,k}$ ,  $\dot{y}_k$  with  $\dot{x}_k$ ,  $u_{2,k}$  with  $u_{3,k}$ , and  $J_{xx}$  with  $J_{yy}$ .

$$A_{3,1} = -\Delta t a_5 g_{2,k}, \quad A_{3,3} = -\frac{\Delta t a_7 l}{J_{yy}}, \quad (71)$$

giving  $D_{3,1} = \Delta t a_5 g_{2,k}$  and  $D_{3,3} = \Delta t a_7 l/J_{yy}$ .

The remaining entries of  $A$  (and hence  $D$ ) are zero, as thrust does not appear in the yaw dynamics ( $A_{4,1} = 0$ ), and roll moment neither appears in  $\Delta s_\theta$  (orthogonal underactuated channel) nor in  $\Delta s_z, \Delta s_\psi$  (fully-actuated channels). The same reasoning eliminates the remaining off-diagonals in columns 2–4, using  $J_r \approx 0$ ; with rotor spin inertia retained, the gyroscopic terms couple the yaw command into rows 2–3 through  $\Omega_r$ . Only column 1 (thrust) and the four

diagonal elements are nonzero. Collecting and applying  $D = -A$ , with rows ordered  $\{s_z, s_\phi, s_\theta, s_\psi\}$  and columns  $\{u_T, u_{M_x}, u_{M_y}, u_{M_z}\}$ ,

$$D_k = \Delta t \begin{bmatrix} \frac{c \phi_k c \theta_k}{m} & 0 & 0 & 0 \\ a_1 g_{1,k} & \frac{a_3 l}{J_{xx}} & 0 & 0 \\ a_5 g_{2,k} & 0 & \frac{a_7 l}{J_{yy}} & 0 \\ 0 & 0 & 0 & \frac{1}{J_{zz}} \end{bmatrix}. \quad (72)$$

Diagonal elements are positive:  $D_{2,2}$  and  $D_{3,3}$  since  $a_3 = a_7 > 0$ ,  $D_{4,4} = \Delta t/J_{zz} > 0$  unconditionally, and  $D_{1,1} = \Delta t c \phi_k c \theta_k / m > 0$  provided  $|\phi_k|, |\theta_k| < 90^\circ$ , an attitude condition that we fold into the explicit trajectory envelope of Section VI.F. At hover,  $D_{2,1}$  and  $D_{3,1} \rightarrow 0$ , leaving  $D_k$  diagonal. Because  $a_1, a_2, a_5, a_6$  depend only on the commanded  $u_T$  and the attitude,  $D_{1,1}$  and  $D_{4,4}$  are computed from state only, and  $u_T$  from  $\tilde{s}_z = s_z - D_{1,1} \xi_1$ , then  $a_1, a_2, a_5, a_6$  from  $u_T$ , then  $D_{2,1}, D_{2,2}, D_{3,1}, D_{3,3}$  to complete the matrix. With  $D_{1,1} > 0$ , a positive thrust deficit ( $\xi_T > 0$ , upper-bound saturation on  $u_T$ ) reduces  $\tilde{s}_z = s_z - D_{1,1} \xi_T$  below  $s_z$ , which in turn reduces commanded  $u_T$ . The symmetric statement holds for lower bound saturation and for the other moment channels. Therefore, we verify that the saturation is being applied in the correct direction.

### E. Schur Stability of the Augmented System

Deriving now the augmented closed-loop system with  $(\tilde{s}, \xi)$ , we aim to establish Schur (discrete-time Hurwitz) stability. The exponential reaching laws for attitude and altitude enforce  $\Delta s_k = -E s_k$ , and so substituting  $s \mapsto \tilde{s}$  in the reaching-law term carries the same identity forward into  $\tilde{s}$ :

$$\Delta s_k|_{\text{commanded}} = -E \tilde{s}_k, \quad E \triangleq \Delta t \text{diag}(\eta_T, \eta_{M_x}, \eta_{M_y}, \eta_{M_z}). \quad (73)$$

As in the unconstrained law, this identity holds inside the clip band  $|s_{z,k}|, |s_{\psi,k}| \leq 1.578$  (Table 4); outside it, the constant-rate reaching step drives the fully-actuated variables into the band, and the certificates below apply once inside. The actually-applied input is  $\bar{u}_k = u_k - \Delta u_k$ . By Equation 57,

$$\Delta s_k = -E \tilde{s}_k + D_k \Delta u_k. \quad (74)$$

The control deficit enters into  $s$  via  $D_k$ . However, it does not enter into  $\tilde{s}$ , as shown below. Combining Equations 51, 50, and 74 with  $X_k \triangleq (\tilde{s}_k^T, \xi_k^T)^T \in \mathbb{R}^8$ ,

$$X_{k+1} = \mathcal{A}_k X_k + \mathcal{B} \Delta u_k, \quad \mathcal{A}_k = \begin{bmatrix} I - E & D_k K_\xi \\ 0 & I - K_\xi \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} 0 \\ I \end{bmatrix}. \quad (75)$$

Two features of  $\mathcal{A}_k$  drive the analysis: (i) the lower-left zero block, so that the deficit does not appear in the  $\tilde{s}$  row, and (ii) the upper-right block  $D_k K_\xi$ , the residual cross-coupling, which decays at the  $\xi$ -rate.

To check the zero block in  $\mathcal{A}_k$ , we expand  $\tilde{s}_{k+1}$  and take  $D_{k+1} \approx D_k$ :

$$\begin{aligned} \tilde{s}_{k+1} &= s_{k+1} - D_k \xi_{k+1} && \text{(Eq. 51)} \\ &= s_k + \Delta s_k - D_k \xi_{k+1} && \text{(unrolling } s_{k+1}) \\ &= s_k + (-E \tilde{s}_k + D_k \Delta u_k) && \\ &\quad - D_k [(I - K_\xi) \xi_k + \Delta u_k] && \text{(Eqs. 74, 50)} \\ &= (s_k - D_k \xi_k) - E \tilde{s}_k + D_k K_\xi \xi_k && (D_k \Delta u_k \text{ cancels)} \\ &= (I - E) \tilde{s}_k + D_k K_\xi \xi_k. && \text{(definition of } \tilde{s}_k) \end{aligned} \quad (76)$$

The two  $D_k \Delta u_k$  contributions (one from the saturated plant, one from the auxiliary update) have opposite signs and cancel out. So, the deficit only enters the  $\xi$  equation, and  $\Delta u_k$  is decoupled from  $\tilde{s}$ . This decoupling permits the diagonal Lyapunov decomposition of Section VI.F.

The matrix  $\mathcal{A}_k$  is block upper-triangular, so  $\sigma(\mathcal{A}_k) = \sigma(I - E) \cup \sigma(I - K_\xi)$ , and both blocks are diagonal:

$$\sigma(\mathcal{A}_k) = \{1 - \eta_\alpha \Delta t\}_\alpha \cup \{1 - k_{\xi \alpha}\}_\alpha. \quad (77)$$

Schur stability ( $|\lambda| < 1$ ) simply becomes:

$$0 < \eta_\alpha \Delta t < 2, \quad 0 < k_\xi \alpha < 2, \quad \forall \alpha. \quad (78)$$

Plant inertia parameters ( $m, J_{xx}, J_{yy}, J_{zz}$ ) enter only through  $D_k$ , which lives in the off-diagonal block and does not affect the spectrum, so the Schur condition is parameter-uniform. The closed-loop decay constants are not: they depend on  $M_D \triangleq \sup_k |D_k|$  through the ISS decay rate  $\nu$  of Section VI.F.

Two steps above are implicit and deserve explicit justification, since for a general linear time-varying (LTV) system, pointwise Schur eigenvalues do not imply stability: the cancellation in Equation 76 takes  $D_{k+1} \approx D_k$ , and the spectrum in Equation 77 is that of the frozen-time matrix  $\mathcal{A}_k$ . Both are rescued by the structure of  $\mathcal{A}_k$ . Because the diagonal blocks are constant and  $D_k$  enters only the off-diagonal block, the frozen-time spectrum is  $k$ -independent, and the LTV system is exponentially stable whenever  $M_D < \infty$ :  $\xi$  decays autonomously through the constant block  $I - K_\xi$ , and  $\tilde{s}$  obeys a constant-coefficient stable recursion driven by the bounded input  $D_k K_\xi \xi_k$ , plus the remainder  $(D_k - D_{k+1}) \xi_{k+1}$  from the  $D_{k+1} \approx D_k$  step. Bounding this term needs the step-to-step variation of  $D_k$ , not just its magnitude  $M_D$ , so we add to the envelope of Equation 95 a slew bound  $|D_{k+1} - D_k| \leq L \Delta t$  along closed-loop trajectories. This holds whenever the thrust and attitude rates that drive  $D_k$  are bounded, which the reaching transient satisfies once the fully-actuated variables enter the clip band. Under it the remainder is  $O(L \Delta t |\xi_k|)$  and vanishes with  $\xi$ . The ISS-Lyapunov inequality of Section VI.F is the formal certificate: with an active deficit the remainder is  $O(|\xi_k| + |\Delta u_k|)$ , carried as an additional input in Equation 91, and the slew bound makes its coefficient  $O(L \Delta t)$ , perturbing  $\nu$  and  $\gamma$  only at higher order. The frozen-time spectrum is therefore diagnostic; the ISS inequality, under Equation 95 and this slew bound, is the operative certificate.

## F. ISS-Lyapunov Stability

Take the candidate Lyapunov function

$$V_k = \tilde{s}_k^T W_s \tilde{s}_k + \rho \xi_k^T W_\xi \xi_k, \quad W_s, W_\xi \succ 0 \text{ diagonal}, \rho > 0, \quad (79)$$

and compute  $\Delta V_k \triangleq V_{k+1} - V_k$  by substituting the closed-loop state updates from (Equations 76 and 50) into  $V_{k+1}$ . The construction follows the discrete-time ISS-Lyapunov framework of Jiang and Wang [14].

- 1) Substituting  $\tilde{s}_{k+1} = (I - E) \tilde{s}_k + D_k K_\xi \xi_k$  from Equation 76 into the first quadratic term of  $V_{k+1}$ ,

$$\begin{aligned} \tilde{s}_{k+1}^T W_s \tilde{s}_{k+1} &= \tilde{s}_k^T (I - E)^T W_s (I - E) \tilde{s}_k \\ &\quad + 2 \tilde{s}_k^T (I - E)^T W_s D_k K_\xi \xi_k \\ &\quad + \xi_k^T K_\xi^T D_k^T W_s D_k K_\xi \xi_k. \end{aligned} \quad (80)$$

- 2) Substituting  $\xi_{k+1} = (I - K_\xi) \xi_k + \Delta u_k$  into the second quadratic term,

$$\begin{aligned} \rho \xi_{k+1}^T W_\xi \xi_{k+1} &= \rho \xi_k^T (I - K_\xi)^T W_\xi (I - K_\xi) \xi_k \\ &\quad + 2 \rho \xi_k^T (I - K_\xi)^T W_\xi \Delta u_k \\ &\quad + \rho \Delta u_k^T W_\xi \Delta u_k. \end{aligned} \quad (81)$$

- 3) Combining the two expansions and subtracting  $V_k$ ,

$$\begin{aligned} \Delta V_k &= \underbrace{\tilde{s}_k^T ((I - E)^T W_s (I - E) - W_s) \tilde{s}_k}_{\text{(a) } \tilde{s}\text{-decay}} \\ &\quad + \underbrace{2 \tilde{s}_k^T (I - E)^T W_s D_k K_\xi \xi_k}_{\text{(b) } \tilde{s}\text{-}\xi \text{ cross}} \\ &\quad + \underbrace{\xi_k^T K_\xi^T D_k^T W_s D_k K_\xi \xi_k}_{\text{(c) cross-square residual}} \\ &\quad + \underbrace{\rho \xi_k^T ((I - K_\xi)^T W_\xi (I - K_\xi) - W_\xi) \xi_k}_{\text{(d) } \xi\text{-decay}} \\ &\quad + \underbrace{2 \rho \xi_k^T (I - K_\xi)^T W_\xi \Delta u_k}_{\text{(e) } \xi\text{-}\Delta u \text{ cross}} \\ &\quad + \underbrace{\rho \Delta u_k^T W_\xi \Delta u_k}_{\text{(f) deficit penalty}}. \end{aligned} \quad (82)$$

- 4) Using  $1 - (1 - x)^2 = x(2 - x)$ , the decay terms (a) and (d) become

$$(I - E)^T W_s (I - E) - W_s = -\text{diag}(\eta_\alpha \Delta t (2 - \eta_\alpha \Delta t) (W_s)_{\alpha\alpha}) < 0, \quad (83)$$

negative definite under Equation 78; the decay rate is maximized at  $\eta_\alpha \Delta t = 1$  and vanishes at  $\eta_\alpha \Delta t \in \{0, 2\}$ . We can also write,

$$(I - K_\xi)^T W_\xi (I - K_\xi) - W_\xi = -\text{diag}(k_{\xi\alpha} (2 - k_{\xi\alpha}) (W_\xi)_{\alpha\alpha}) < 0. \quad (84)$$

- 5) Apply Young's inequality to absorb the  $\tilde{s}$ - $\xi$  cross-term (b). For any  $\mu > 0$  and positive-definite  $P$ ,

$$2u^T P v \leq \mu u^T P u + \frac{1}{\mu} v^T P v. \quad (85)$$

Apply with  $u = (I - E)\tilde{s}_k$ ,  $P = W_s$ ,  $v = D_k K_\xi \xi_k$ ,

$$(b) \leq \mu \tilde{s}_k^T (I - E)^T W_s (I - E) \tilde{s}_k + \frac{1}{\mu} \xi_k^T K_\xi^T D_k^T W_s D_k K_\xi \xi_k. \quad (86)$$

The first Young term carries the quadratic form  $(I - E)^T W_s (I - E)$ , not the decay term (a), so its per-channel coefficient is  $+\mu (1 - \eta_\alpha \Delta t)^2 (W_s)_{\alpha\alpha}$ . Combining it with (a) gives the net diagonal coefficient  $-\left[\eta_\alpha \Delta t (2 - \eta_\alpha \Delta t) - \mu (1 - \eta_\alpha \Delta t)^2\right] (W_s)_{\alpha\alpha}$  on each  $\tilde{s}_\alpha^2$ , which is negative definite iff the bracket is positive on every channel:

$$0 < \mu < \min_\alpha \frac{\eta_\alpha \Delta t (2 - \eta_\alpha \Delta t)}{(1 - \eta_\alpha \Delta t)^2}. \quad (87)$$

Hence  $\mu \in (0, 1)$  alone is insufficient: the absorbed form scales as  $(1 - \eta_\alpha \Delta t)^2$ , not  $\eta_\alpha \Delta t (2 - \eta_\alpha \Delta t)$ . For the saturated formulation's gains of Table 4 at 50 Hz ( $\eta_z \Delta t = 0.146$ ,  $\eta_\psi \Delta t = 0.166$ ,  $\eta_\phi \Delta t = \eta_\theta \Delta t = 0.100$ ), the binding  $(\phi, \theta)$  channels require  $\mu < 0.234$ . The second Young term adds to the cross-square residual (c).

- 6) Apply Young's inequality once more, now to absorb the  $\xi$ - $\Delta u$  cross-term (e), with  $u = (I - K_\xi)\xi_k$ ,  $v = \Delta u_k$ ,  $P = \rho W_\xi$ , parameter  $\nu_2 > 0$ ,

$$(e) \leq \rho \nu_2 \xi_k^T (I - K_\xi)^T W_\xi (I - K_\xi) \xi_k + \frac{\rho}{\nu_2} \Delta u_k^T W_\xi \Delta u_k. \quad (88)$$

The first piece carries the form  $(I - K_\xi)^T W_\xi (I - K_\xi)$ , contributing  $+\rho \nu_2 (1 - k_{\xi\alpha})^2 (W_\xi)_{\alpha\alpha}$  per channel; as with  $\mu$  in Equation 87, this erodes (rather than cleanly rescales by  $1 - \nu_2$ ) the  $\xi$ -decay margin (d), and the  $\xi$ -channel stays negative definite only for

$$0 < \nu_2 < \min_\alpha \frac{k_{\xi\alpha} (2 - k_{\xi\alpha})}{(1 - k_{\xi\alpha})^2}. \quad (89)$$

Under the matched gains of Equation 52,  $k_{\xi\alpha} = \eta_\alpha \Delta t$  and this bound coincides with Equation 87. The implemented 1.2741 scaling separates the two: at 50 Hz it gives  $\nu_2 < 0.313$  against  $\mu < 0.234$ , both binding on the  $(\phi, \theta)$  channels. The second piece adds to the deficit penalty (f). We fix  $\mu$  and  $\nu_2$  within Equations 87 and 89 before selecting  $\rho$  in the next step, because the domination condition there must hold against the  $\xi$ -decay margin that survives this erosion.

- 7) With  $\mu$  and  $\nu_2$  fixed, and provided  $M_D = \sup_k |D_k| < \infty$  (assumption (iii) below), the sum of the cross-square residual (c) and the leftover from the first Young absorption (Equation 86) is bounded by  $\left[(1 + 1/\mu) M_D^2 |K_\xi|^2 |W_s| + |\xi_k|^2\right]$ . We then choose  $\rho$  large enough that the eroded  $\xi$ -decay dominates this bound,

$$\rho \min_\alpha \left[ k_{\xi\alpha} (2 - k_{\xi\alpha}) - \nu_2 (1 - k_{\xi\alpha})^2 \right] (W_\xi)_{\alpha\alpha} > (1 + \frac{1}{\mu}) M_D^2 |K_\xi|^2 |W_s|, \quad (90)$$

which is achievable for any finite  $M_D$ , since Equation 89 keeps the bracket positive on every channel. The combined contribution of terms (a)–(e), after both absorptions, is then negative definite in  $(\tilde{s}, \xi)$ . Collecting,

$$\Delta V_k \leq -\nu V_k + \gamma |\Delta u_k|^2, \quad (91)$$

where

$$\nu = \frac{1}{\max(|W_s|, \rho |W_\xi|)} \min_\alpha \left[ \eta_\alpha \Delta t (2 - \eta_\alpha \Delta t) - \mu (1 - \eta_\alpha \Delta t)^2 \right] (W_s)_{\alpha\alpha},$$

$$\min_{\alpha} \rho \left[ k_{\xi} \alpha (2 - k_{\xi} \alpha) - \nu_2 (1 - k_{\xi} \alpha)^2 \right] (W_{\xi})_{\alpha\alpha} - \left(1 + \frac{1}{\mu}\right) M_D^2 |K_{\xi}|^2 |W_s|, \quad (92)$$

$$\gamma = \rho \left(1 + \frac{1}{\nu_2}\right) |W_{\xi}|. \quad (93)$$

These constants are exact rather than order-of-magnitude: the first branch of the minimum is the per-channel  $\tilde{s}$ -decay that survives the first Young absorption, the second is the eroded  $\xi$ -decay minus the absorbed cross-square bound, and so  $\nu > 0$  holds precisely when Equations 87, 89, and 90 do. Equation 91 is the discrete-time ISS-Lyapunov inequality in the standard form (cf. Jiang and Wang [14], Definition 3.2 and Example 3.4) used throughout the saturated-systems literature.

Under (i) the Schur condition (Equation 78), (ii) hover-trim feasibility  $u_{\text{hover}} \in \text{int}(\mathcal{U})$ , and (iii)  $M_D = \sup_k |D_k| < \infty$ , the closed-loop system satisfies:

- 1) If  $\Delta u_k = 0$  for all  $k$ , then  $V_k \leq (1 - \nu)^k V_0$  and  $(\tilde{s}_k, \xi_k) \rightarrow 0$  exponentially.
- 2) If  $\Delta u_k \rightarrow 0$ , then  $V_k \rightarrow 0$  and  $(\tilde{s}_k, \xi_k) \rightarrow 0$  asymptotically; the controller returns to the nominal dSMC as the control deficit decreases.
- 3) If  $\sup_k |\Delta u_k|^2 \leq D_{\text{sup}}$ , then  $\limsup_k V_k \leq \gamma D_{\text{sup}}/\nu$ , i.e., the controller (and the system response) is ultimately bounded.

Assumptions (i) and (ii) are directly checkable from the gains and the feasible set. Assumption (iii) is not: it is nontrivial precisely in the tumbling regime this paper targets, and we state it as an explicit trajectory envelope rather than leaving it implicit. From Table 5,

$$D_{2,1} = \Delta t a_1 g_{1,k} = \frac{5.1762 \Delta t T_{y,k}}{u_T c \psi_k}, \quad D_{3,1} = \Delta t a_5 g_{2,k} = -\frac{5.1762 \Delta t T_{x,k}}{u_T c \phi_k c \psi_k}, \quad (94)$$

both of which grow without bound as the commanded  $u_T \rightarrow 0$  or as  $c \psi_k \rightarrow 0$  (respectively  $c \phi_k c \psi_k \rightarrow 0$ ), and a ballistically launched, tumbling vehicle can pass near both. We therefore require: there exist  $\epsilon_u, \epsilon_a > 0$  such that, along closed-loop trajectories,

$$|u_T| \geq \epsilon_u, \quad |c \psi_k| \geq \epsilon_a, \quad |c \phi_k c \psi_k| \geq \epsilon_a, \quad |\phi_k|, |\theta_k| < 90^\circ, \quad (95)$$

where the first three conditions bound the cross-coupling magnitudes and the last keeps  $D_{1,1} > 0$ , as used by the deficit-direction argument of Section VI.D. Under Equation 95, every nonzero entry of  $D_k$  is bounded— $|T_{x,k}|, |T_{y,k}| \leq 1$ , since each is the inner product of  $(c \phi_k s \theta_k, s \phi_k)$ , of norm at most one, with a unit vector—and bounding the spectral norm by the Frobenius norm over the six nonzero entries gives the explicit certificate

$$M_D \leq \sqrt{6} \Delta t \max \left( \frac{1}{m}, \frac{5.1762}{\epsilon_u \epsilon_a}, \frac{a_3 l}{J_{xx}}, \frac{a_7 l}{J_{yy}}, \frac{1}{J_{zz}} \right) < \infty. \quad (96)$$

For the successful-landing trials, Figure 5 shows the applied effort norm holding at the allocation bound through the reaching transient and then relaxing toward hover trim, never approaching zero thrust—evidence for the nonzero-thrust condition of Equation 95, though not for its attitude conditions. This check is also one-sided: the successful trials are, by selection, exactly those the controller recovers, so the figure shows the envelope is non-vacuous on the recovered set but does not establish that failing trials fail because they leave it. Characterizing where failures exit the envelope is left to future work. We also note that the envelope is needed only to bound the cross-coupling magnitude, not the anti-windup direction on the thrust channel: substituting  $D_{1,1} = \Delta t c \phi_k c \theta_k / m$  into the wrapped thrust law cancels the attitude factors, giving  $\partial u_T / \partial \xi_T = -\eta_z \Delta t$ , so a positive thrust deficit reduces the commanded thrust by the same amount at every attitude.

## G. Reduction to the Unconstrained Limit

If the dSMC command  $u_k$  lies inside  $\mathcal{U}$  for all  $k$ , then the saturation collapses and the controller simply returns  $\bar{u}_k = u_k$  and  $\Delta u_k = 0$ , which is the unconstrained dSMC control effort. The auxiliary state update becomes  $\xi_{k+1} = (I - K_{\xi}) \xi_k$ , and so, decays to 0 exponentially as well. Equation 51 thus gives  $\tilde{s}_k \rightarrow s_k$  exponentially, and the control law converges to the baseline dSMC. As  $\xi_k \rightarrow 0$ , the Lyapunov function reduces to

$$V_k \rightarrow s_k^T W_s s_k. \quad (97)$$

With  $W_s = \frac{1}{2} I$  this is  $\frac{1}{2} \sum_{\alpha} s_{\alpha,k}^2$ , the per-channel sum of  $V_{i,k}$  from Equation 28, and clearly still positive definite in  $s$ . So, we recover our unconstrained Lyapunov positivity when the saturation ceases, as expected.

## H. Steady-State Residuals Under Sustained Saturation

The steady-state response to a sustained and constant control deficit  $\Delta u_k = c$  is the fixed point  $X^\infty$  of Equation 75,  $(I - \mathcal{A}) X^\infty = \mathcal{B} c$ , treating  $D_k$  as constant ( $D_k = D$ ) over the steady-state window. When the ideal control effort eventually returns to the feasible set and  $\Delta u_k \rightarrow 0$ , both  $\xi$  and  $\tilde{s}$  decay to zero, and the controller resumes nominal sliding. However, under a sustained  $|c| > 0$ , the controller fails to converge to the sliding surface, and instead maintains some residual error, with  $V^\infty \leq \gamma |c|^2 / \nu$  from the ISS bound (Equation 91), quadratic in the deficit magnitude.

The fixed-point equation

$$\begin{bmatrix} E & -D K_\xi \\ 0 & K_\xi \end{bmatrix} \begin{bmatrix} \tilde{s}^\infty \\ \xi^\infty \end{bmatrix} = \begin{bmatrix} 0 \\ c \end{bmatrix} \quad (98)$$

gives, from the bottom up,

$$\xi^\infty = K_\xi^{-1} c, \quad \tilde{s}^\infty = E^{-1} D c. \quad (99)$$

Under matched gains (Equation 52),  $K_\xi = E$  and

$$\xi_\alpha^\infty = \frac{c_\alpha}{\eta_\alpha \Delta t}, \quad \tilde{s}_i^\infty = \frac{(D)_{ij} c_j}{\eta_i \Delta t}. \quad (100)$$

At the implemented 1.2741 scaling only  $\xi^\infty$  changes, shrinking to  $\xi_\alpha^\infty = c_\alpha / (1.2741 \eta_\alpha \Delta t)$ : by Equation 99,  $\tilde{s}^\infty = E^{-1} D c$  does not involve  $K_\xi$ , so every residual ratio below is unchanged. For a deficit on channel  $\alpha$ ,  $c = c_\alpha e_\alpha$ , the same-channel ratio  $|\tilde{s}_i^\infty|/|c_\alpha| = |D_{i\alpha}|/(\eta_i \Delta t)$  becomes, substituting from Equation 72,

$$\frac{|\tilde{s}_z^\infty|}{|c_T|} = \frac{c_\phi c_\theta}{m \eta_T} \approx \frac{1}{m \eta_T}, \quad \frac{|\tilde{s}_\phi^\infty|}{|c_{M_x}|} = \frac{a_3 l}{J_{xx} \eta_{M_x}}, \quad (101)$$

$$\frac{|\tilde{s}_\theta^\infty|}{|c_{M_y}|} = \frac{a_7 l}{J_{yy} \eta_{M_y}}, \quad \frac{|\tilde{s}_\psi^\infty|}{|c_{M_z}|} = \frac{1}{J_{zz} \eta_{M_z}}, \quad (102)$$

which are  $\Delta t$ -independent. A pure thrust deficit  $c = c_T e_1$  simultaneously perturbs the underactuated channels via column 1 of  $D$ :

$$\frac{|\tilde{s}_\phi^\infty|}{|c_T|} = \frac{|a_1| |g_{1,k}|}{\eta_{M_x}}, \quad \frac{|\tilde{s}_\theta^\infty|}{|c_T|} = \frac{|a_5| |g_{2,k}|}{\eta_{M_y}}. \quad (103)$$

These same-channel ratios are monotone decreasing in  $\eta_\alpha$ , so their infimum over the Schur-stable interval  $\eta_\alpha \Delta t \in (0, 2)$  (equivalently, the discrete-time sliding-mode existence condition of Sarpturk et al. [15]) is approached only as  $\eta_\alpha \Delta t \rightarrow 2$  (the Schur boundary, where the decay rate  $\eta_\alpha \Delta t (2 - \eta_\alpha \Delta t)$  vanishes, giving  $\Delta t / (2m)$  for thrust. At the decay-rate-optimal operating point  $\eta_\alpha \Delta t = 1$  the ratio is instead  $\Delta t / m$  for thrust and  $a_3 l \Delta t / J_{xx}$  for roll (identically for pitch;  $\Delta t / J_{zz}$  for yaw), so the gain choice trades residual size against convergence speed. The moment-vs-thrust residual ratio scales as

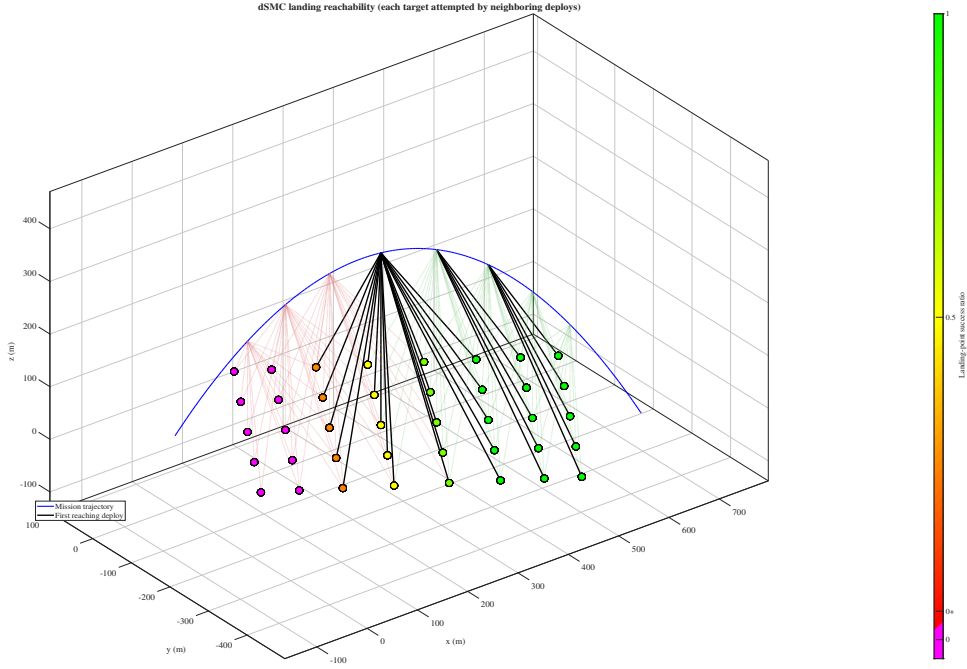
$$\frac{|\tilde{s}_\phi^\infty|/|c_{M_x}|}{|\tilde{s}_z^\infty|/|c_T|} = \frac{m a_3 l}{J_{xx}} \cdot \frac{\eta_T}{\eta_{M_x}} = \frac{a_3}{l} \cdot \frac{m l^2}{J_{xx}} \cdot \frac{\eta_T}{\eta_{M_x}}, \quad (104)$$

in which the vehicle dependence is carried by the dimensionless mass-distribution factor  $m l^2 / J_{xx}$ , equal to 17.8 for this vehicle. The ratio is therefore large whenever  $m l^2 \gg J_{xx}$ , the thin rotor-plane mass distribution typical of small quadrotors, which is a trait of the underactuated system and not of the saturation handler. Recovery to the sliding surface, therefore, requires the saturation deficit itself to eventually disappear; the controller alone cannot drive the residual to zero while the actuator remains saturated, as the system is unable to execute the commanded dSMC control efforts.

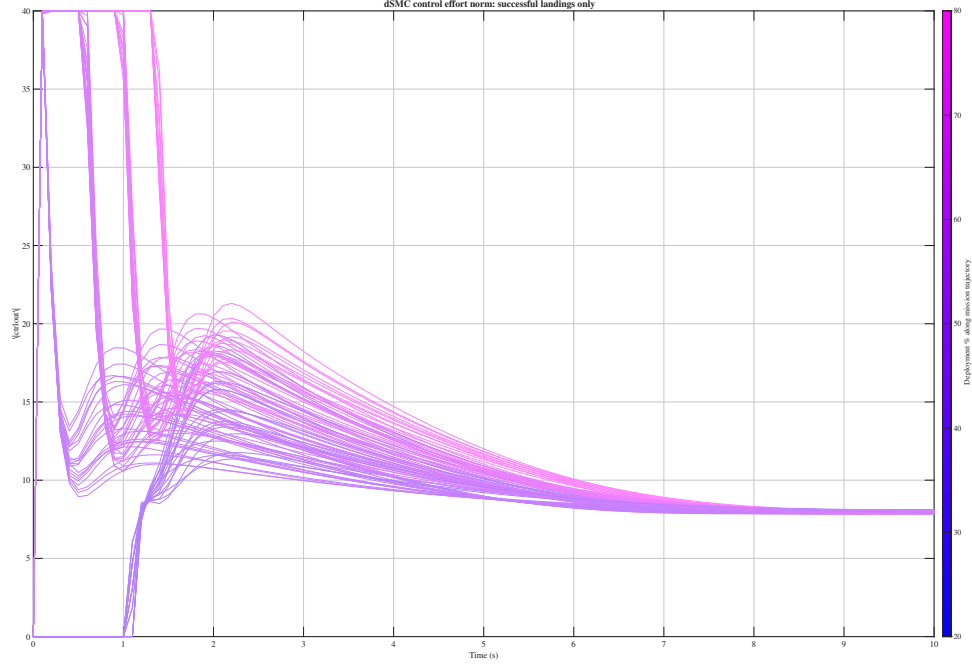
## I. Saturated Control Results

We tested the performance of the saturated control law, against the baseline dSMC with a simple per-rotor speed clip. When saturation was enabled, we ran the priority-weighted projection (Equations 46–47) and auxiliary update steps (Equation 50); when saturation was disabled, we left the unconstrained dSMC command alone except for the per-rotor mixer clip in order to respect the physical control constraints of the vehicle. To test different deployment profiles, we tested landing points directly under the ballistic track, as well as at 100 meters and 200 meters to either side of the track. Eight deployment points are sampled uniformly at 20–80% of the trajectory arc length; each landing target (the ground projection of a deployment point plus the cross-track offset) is attempted from the neighboring deployment points at index offsets  $\{-1, 0, +1, +2\}$ , giving 140 valid trials of the  $8 \times 5 \times 4$  grid after boundary exclusion, each with a 60 s stabilization window. We evaluate successful landings according to the criterion in Equation 40.

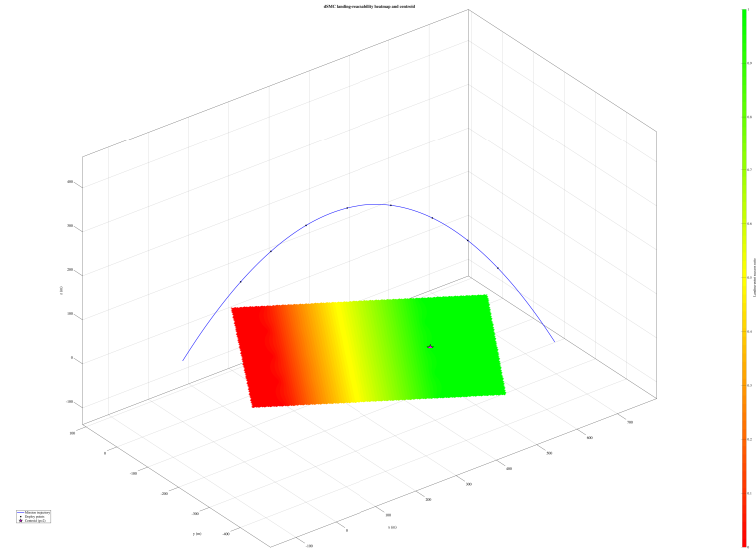
Figure 4 shows the per-target landing envelope under saturation. Targets across most of the central band are reached by at least one neighboring deploy; the misses concentrate at the short-range and lateral edges of the swept footprint. Figure 5 shows why the controller stays inside this envelope, as the applied control-effort norm (the post-allocation command  $\bar{u}_k$ ) sits at the allocation bound ( $|\bar{u}_k| \approx 40 = 4b \Omega_{\max}^2$ , the maximum command norm over  $\mathcal{U}$ , attained only at full collective thrust) for the first 1–2 s of the reaching transient, then relaxes toward hover trim. As discussed in Sections VI.F and VI.H, prolonged control saturation forfeits convergence to the sliding surface—only the ultimate-boundedness bound of Equation 91 survives—and in our simulation the resulting residuals are large enough that the vehicle crashes. Under the unconstrained controller, the same norm is far larger: a median per-trial peak of  $\approx 490$  ( $12\times$  the applied bound), diverging past  $10^9$  on the steepest deploys as the commanded thrust approaches zero; under per-rotor clipping the norm is capped at the same bound of 40, but without the priority structure or the anti-windup state the vehicle destabilizes, and only 31 of 140 clipped trials land, versus 85 of 140 under the saturation handler (Table 7). Figure 6 aggregates the per-target data into a continuous reachability surface and locates the power-weighted centroid of reachability, which is essentially the most reachable point in the space. Table 7 collects the four landing statistics: the reachability fraction (the share of valid sweep trials that land), peak success ratio, half-success radius (the radius at which the radially binned mean success ratio falls to half its peak), and the centroid location, for the saturation handler enabled vs. the clipped baseline.



**Fig. 4** Per-target landing reachability under saturation. Each landing target is attempted by neighboring deploys; markers are colored by the per-target success ratio, and black lines connect each target to its earliest reaching deploy.



**Fig. 5** Per-trial applied-command norm  $|\bar{u}_k|$  during ballistic recovery with the saturation handler enabled, restricted to trials that achieved a successful landing. Traces are colored by deploy point along the mission trajectory.



**Fig. 6** Landing-success ratio over the deploy  $\times$  cross-track sweep with the saturation handler enabled. The interpolated reachability surface is plotted beneath the ballistic mission trajectory, along with the power-weighted reachability centroid ( $p = 2$ , star).



**Table 7 Landing-reachability statistics for the ballistic-recovery sweep with the saturation handler enabled versus the clipped baseline. Centroid uses the power-weighted estimator with  $p = 2$ .**

| Metric                  | Saturation        | Control Clipping Only |
|-------------------------|-------------------|-----------------------|
| Reachability fraction   | 0.607             | 0.221                 |
| Peak success ratio      | 0.972             | 0.500                 |
| Half-success radius (m) | 239.40            | 203.08                |
| Centroid $(x, y)$ (m)   | (420.24, -257.48) | (414.84, -230.05)     |

## VII. Conclusions

The dSMC recovers more of the ballistic-deployment envelope than the LQR, PID, or cSMC controllers, though no controller recovers every sampled state. Under simple motor-speed clipping the sweep reachability is 22.1%; the priority-weighted saturation handler raises it to 60.7%. This matters for launched-UAV operations because, given the smaller actuators of a launched UAV, control effort constraints can be severe. The COTS techniques (LQR and PID) and the continuous sliding mode controller do not stabilize the vehicle at most points along the trajectory - this is consistent with our intuition, as the COTS controllers assume linearity and the cSMC controller assumes continuity. For launched-drone operations, and for the off-nominal initial conditions and aggressive flight tested here, the dSMC formulation recovers more of the deployment envelope than the COTS controllers and, unlike cSMC, keeps its stability certificates under our discretized plant. These certificates cover the four actuated channels, and under saturation they hold while the trajectory stays inside the envelope of Equation 95.

### A. Future Work

With an effective saturated controller and centroid generation logic in place, we want to additionally study the inverse problem of mapping from a desired centroid to the ballistic targeting variables (launch azimuth, elevation, velocity, exit spin, etc) required for a min-time landing, a max-likelihood landing, and a min-effort landing. Currently, we are examining numerical surrogate-based approaches to solving this, and have working prototypes in simulation. We plan to move to flight-test verification after, focusing on the ballistic case. Measuring real moments of inertia and drag parameters is the first step toward closing the sim-to-real gap. Increment-weather operations will require a basic weather model. The dSMC's stability certificates concern the sliding variables and the saturation deficit, not initial-condition or parameter perturbations. Whether the recovered subset (9 of 17 sampled states) also tolerates small velocity and orientation perturbations is an open question we plan to test.

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## Generative AI Statement

The Anthropic Claude [19] generative AI tool was used to improve the formatting and layout of the equations and tables, to reformat Microsoft Word equations to LaTeX formatting, to convert from MATLAB code to LaTeX equations, and to improve general syntax and grammar. The authors assume full responsibility for all results and derivations presented in this paper, and any errors which may be present therein.

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